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Common factors and spatial dependence: an application to US house prices

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ABSTRACT

This article considers panel data models with cross-sectional dependence arising from both spatial autocorrelation and unobserved common factors. It proposes estimation methods that employ cross-sectional averages as factor proxies, including the 2SLS, Best 2SLS, and GMM estimations. The proposed estimators are robust to unknown heteroskedasticity and serial correlation in the disturbances, unrequired to estimate the number of unknown factors, and computationally tractable. The article establishes the asymptotic distributions of these estimators and compares their consistency and efficiency properties. Extensive Monte Carlo experiments lend support to the theoretical findings and demonstrate the satisfactory finite sample performance of the proposed estimators. The empirical section of the article finds strong evidence of spatial dependence of real house price changes across 377 Metropolitan Statistical Areas in the US from 1975Q1 to 2014Q4.

KEYWORDS

Cross-sectional dependence; common factors; spatial panel data models; generalized method of moments; house prices

JEL CLASSIFICATIONS

C13; C23; R21; R31

1. Introduction

The past decade has seen a growing attention to panel data models with cross-sectional dependence, which refers to the interaction between cross-section units such as households, firms, regions, and countries. Researchers have become increasingly aware that ignoring cross-sectional dependence in panel data analysis could lead to inconsistent estimates and misleading inferences. The interdependence among individual units is prevalent in all kinds of economic activities. It could arise from common factors that influence a large number of economic agents, such as technological change and oil price fluctuations. It could also originate from certain explicit correlation structures formed by spatial arrangements, production networks, and social interactions. Accordingly, two main modeling approaches have been proposed to characterize this phenomenon: the common factor models and the spatial econometric models. In the former, cross-sectional dependence is captured by a number of observable or latent factors (or common shocks); in the latter, it is represented by spatial weights matrices typically based on physical, economic, or social distance. Although describing the same phenomenon, these two strands of literature have been developing separately, with different sets of assumptions and emphases. Therefore, efforts are called for to investigate the connections and differences between these two modeling approaches.

This article aims to bring together factor and spatial models for a unified characterization of cross-sectional dependence. The main contributions of the article are twofold. First, it considers a

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joint modeling of the two sources of cross-sectional dependence in panel data models: common factors and spatial interactions. It discusses identification conditions and proposes estimation methods for the joint model. Second, the article provides a detailed empirical application to house price changes in the US and finds strong evidence of spatial effects. The empirical findings are robust and could carry important policy and business implications.

Specifically, our model specifications allow the common effects to be unobservable and the spatial dependence to be an inherent property of the dependent variable. We are faced with two major challenges in devising an estimation strategy. One is related to the unobserved factors, and the other is associated with the endogenous spatial lags of the dependent variable. The estimators developed in this article approximate the unobserved factors by cross-sectional averages of the dependent and independent variables, and then utilize instrumental variables and other moment conditions to resolve the endogeneity problem. These estimators do not require estimating the number of factors, which is well known to be a challenging task. Moreover, they are robust to both heteroskedasticity and serial correlations in the disturbances, and they are computationally attractive. We show that the proposed estimators, including the two-stage least squares (2SLS), Best 2SLS, and generalized method of moments (GMM) estimators, are consistent as long as the cross-section dimension (N) is large, irrespective of the size of the time series dimension (T). Furthermore, they are asymptotically normally distributed without nuisance parameters, provided that T is relatively smaller than N, as both N and T tend jointly toward infinity. We conduct a wide range of Monte Carlo experiments, the results of which are congruent with the theory and show that the proposed estimators have satisfactory finite-sample properties.

The proposed estimation methods are applied in order to analyze changes in real house price the US across 377 Metropolitan Statistical Areas (MSAs) from 1975Q1 to 2014Q4. The study demonstrates the importance of the effective removal of common effects in evaluating the strength of spatial connections. It documents significant spatial dependence in house price changes. It also shows that population and income growth significantly increase house price growth through both direct effect and spillover effect. These findings are fairly robust to various specifications of the spatial weights, including weights based on distance, on migration flows, and on pairwise correlations of the de-factored observations.

The theoretical analysis in this article belongs to a recent and growing literature on panel data models with cross-sectional dependence (CSD). Chudik, Pesaran, and Tosetti (2011) introduce the notions of weak and strong CSD. Applying these concepts, a spatial model can be shown to be a form of weak CSD, whereas the standard factor model represents a form of strong CSD (Bailey, Holly, and Pesaran, 2016; Pesaran and Tosetti, 2011). Bailey, Kapetanios, and Pesaran (2016) propose measuring the degree of CSD by an exponent of dependence, which captures how fast the variance of the cross-sectional average declines with the cross-section dimension, N. Using this exponent of cross-sectional dependence, Pesaran (2015) further discusses testing for weak CSD in large panels.[1]

The characterization of CSD is divided into two areas of writing. On the one hand, there is a large body of literature on common factor models. Recent contributions on large panel data models with common factors include Pesaran (2006), Bai (2009), Bai and Li (2012), and Moon and Weidner (2017), just to name a few. Our study is particularly related to an influential paper by Pesaran (2006), who develops common correlated effects (CCE) estimators for panel data models with multifactor error structure. The basic idea behind the CCE estimators is to filter the unobserved factors with cross-sectional averages. In follow-up studies, Kapetanios, Pesaran, and Yamagata (2011) show that the CCE estimators are still applicable if the unobserved factors

[1]For overviews of the literature on panel data models with error cross-sectional dependence, see Sarafidis and Wansbeek (2012) and Chudik and Pesaran (2015b).

follow unit root processes; Chudik and Pesaran (2015a) extend the estimation approach to models with lagged dependent variables and weakly exogenous regressors.

On the other hand, the present article also draws from the spatial econometrics literature.[2] Two main classes of methods have been developed to estimate spatial models: the maximum likelihood (ML) techniques (Anselin, 1988; Aquaro, Bailey, and Pesaran, 2019; Lee, 2004; Lee and Yu, 2010a; Yu, Jong, and Lee, 2008), and the instrumental variables (IV)/GMM approaches (Kelejian and Prucha, 1999, 2010; Lee, 2007; Lee and Yu, 2014; Lin and Lee, 2010). The estimation strategy in the current article is related to and builds on the GMM framework. Regarding the identification conditions of spatial models, a systematic discussion is provided in a recent study by Lee and Yu (2016) under the assumption that the sample size is finite. Aquaro, Bailey, and Pesaran (2019) also conduct a detailed investigation of the identifiability of spatial models with heterogeneous coefficients. The present article sheds light on the identification of spatial models with factors.

The current article is most closely related to a number of more recent studies that consider both common factors and spatial effects. Pesaran and Tosetti (2011) consider models where the idiosyncratic errors are spatially correlated and subject to common shocks. Bai and Li (2014) specify the spatial autocorrelation on the dependent variable while assuming the presence of unobserved common shocks. They advocate a pseudo-ML method that simultaneously estimates a large group of parameters, including the heterogeneous factor loadings and heterogeneous variances of the disturbances. A similar approach is considered by Bai and Li (2015) for dynamic models. Other studies within the ML framework include Shi and Lee (2017) and Lu (2017). However, besides computational complexities, the ML methods are not robust to serial correlation in the errors, and they require knowing or estimating the number of latent factors.[3] Instead of estimating the two effects jointly, Bailey, Holly, and Pesaran (2016) propose a two-stage approach that extracts the common factors in the first stage and then estimates the spatial connections in the second stage. Nonetheless, a formal distribution theory that takes into account the first-stage sampling errors is not yet available.

The empirical investigation in the present article is concerned with the spatial dependence in house prices. The phenomenon that house price variations tend to exhibit spatial correlations has received increasing attention from economists over the past two decades, although little consensus has been reached regarding the spatial transmission mechanism. Possible explanations include migration, equity transfer, spatial arbitrage, and spatial patterns in the determinants of house prices (Meen, 1999). Researchers have obtained evidence on the spatial spillovers of house prices in the US at different levels of aggregation using various methods. For example, Pollakowski and Ray (1997) examine nine US Census divisions as well as the New York metropolitan area using a vector autoregressive model. Brady (2011) focuses on the diffusion of house prices across a panel of California counties by means of impulse response functions. Holly, Pesaran, and Yamagata (2010) analyze US house prices at the State level using a spatial error model, where the importance of spatial effects is evaluated by fitting a spatial model to the residuals from a CCE estimation procedure. Brady (2014) also considers State level house prices but utilizes spatial impulse response functions from a single-equation spatial autoregressive model. The current article focuses on the extent to which house prices are interdependent among near 400 MSAs in the US. Little research has investigated this issue at the MSA level. One exception is the study undertaken by Cohen, Ioannides, and Thanapisitikul (2016), who incorporate geography into an autoregressive model via cross-lag effects and do not employ a spatial econometric approach.[4] Our

[2]Comprehensive reviews of spatial econometrics can be found in books including Anselin (1988) and Elhorst (2014). Also see the survey article by Lee and Yu (2010b) for the latest developments in spatial panel data models.

[3]Much is written on estimating the number of unobservable factors. See, for example, Bai and Ng (2002, 2007), Kapetanios (2010), and Stock and Watson (2011).

[4]Cohen, Ioannides, and Thanapisitikul (2016) also use a house price index different from ours. Specifically, the authors adopt the consolidated house price index by the Office of Federal Housing Enterprise Oversight that covers 363 MSAs over the period of 1996–2013.

empirical analysis is closely related to the inquiry by Bailey, Holly, and Pesaran (2016), who examine MSA level house price changes with a two-stage procedure. In comparison, besides using more recent data on updated MSA delineations, the present article adopts a different estimation approach that jointly considers common factors and spatial dependence. It also explores the direct and indirect effects of possible determinant variables on house price growth, and considers the specification of spatial weights matrix based on migration flows.

The rest of the article is organized as follows. Section 2 specifies the model and describes the idea of approximating the unobserved factors with cross-sectional averages. Section 3 establishes the asymptotic distributions of the 2SLS, Best 2SLS, and GMM estimators. Section 4 reports the results of Monte Carlo experiments. Section 5 presents an empirical application to US house prices, and finally, Section 6 concludes. Appendix A summarizes frequently used notations and variable definitions. Appendix B contains proofs of the main theorems. Appendix C gives further details on data sources and variable transformations. The Supporting Information provides a list of lemmas used in the main proofs, and additional results of Monte Carlo simulations and empirical findings.

2. The model and assumptions

Consider the following spatial autoregressive (SAR) model with common factors,

$$\begin{aligned} y_{it} &= \rho y_{it}^{*} + \boldsymbol{\beta}'\mathbf{x}_{it} + \boldsymbol{\gamma}_i'\mathbf{f}_t + e_{it}, \\ \mathbf{x}_{it} &= \mathbf{A}_i'\mathbf{f}_t + \mathbf{v}_{it}, \end{aligned} \tag{1}$$

for $i = 1, 2, ..., N$, and $t = 1, 2, ..., T$, where y_{it} is the dependent variable of unit i at time t, and $y_{it}^{*} = \sum_{j=1}^{N} w_{ij} y_{jt}$, which represents the endogenous interaction effects (or spatial lag effects) among the dependent variable. The matrix $\mathbf{W} = (w_{ij})_{N\times N}$ is a specified spatial weights matrix of known constants. It characterizes neighborhood relations, which are typically based on a geographical arrangement or on socio-economic connections of the cross-section units.[5] The parameter ρ captures the strength of spatial dependence across observations on the dependent variable and is known as the spatial autoregressive coefficient. The $k \times 1$ vector $\mathbf{x}_{it} = (x_{it,1}, x_{it,2}, ..., x_{it,k})'$ contains individual-specific explanatory variables, and $\boldsymbol{\beta}$ is the corresponding vector of coefficients, where k is assumed to be a known fixed number. The variables e_{it} and $\mathbf{v}_{it} = (v_{it,1}, v_{it,2}, ..., v_{it,k})'$ are the idiosyncratic disturbances associated with y_{it} and $\mathbf{x}_{it}$ processes, respectively. The $m \times 1$ vector $\mathbf{f}_t = (f_{1t}, f_{2t}, ..., f_{mt})'$ represents unobserved common factors, where m is fixed but possibly unknown. The factor loadings, $\boldsymbol{\gamma}_i$ and $\mathbf{A}_i$, are nonrandom and capture heterogeneous impacts from the common effects on cross-section units.[6] Overall, the term ρy_{it}^{*} captures the spatial effect, while $\boldsymbol{\gamma}_i'\mathbf{f}_t$ captures the common factor effect. The latter is also referred to in the literature as an interactive effect, since it can be viewed as a generalization of the traditional additive fixed effect. The parameters of interest throughout this article are $\boldsymbol{\delta} = (\rho, \boldsymbol{\beta}')'$.

In model (1), the explanatory variables are specified so that they can be influenced by the same factors that affect the dependent variable. Such a specification is reasonable in practice and has been considered in studies including Pesaran (2006) and Bai and Li (2014). Also note that this model can be readily extended without additional complication to include observable factors such as intercepts, seasonal dummies, and deterministic trends;[7] here, we focus on unobservable factors for expositional clarity.

[5] In general, the spatial weights and variables in the model may dependent on N, but we suppress subscript N throughout the article for ease of notation.

[6] The heterogeneity in factor loadings may arise, for example, from differences in endowment, technical rigidities, or innate ability.

[7] See Remark 2 of Pesaran (2006).

To cope with the unknown factors in model (1), we replace them with cross-sectional averages of the dependent and individual-specific independent variables, following the idea pioneered by Pesaran (2006). To see why this approximation works for the SAR model, we begin by rewriting model (1) as follows:

$$\begin{pmatrix} y_{it} - \rho\sum_{j=1}^{N} w_{ij}y_{jt} - \boldsymbol{\beta}'\mathbf{x}_{it} \\ \mathbf{x}_{it} \end{pmatrix} = \boldsymbol{\Phi}_i'\mathbf{f}_t + \mathbf{u}_{it}, \tag{2}$$

where $\boldsymbol{\Phi}_i = (\gamma_i, \mathbf{A}_i)$, $\mathbf{u}_{it} = (e_{it}, \mathbf{v}_{it}')'$. Then, stacking (2) by individual unit for each time period, the model can be expressed more compactly as

$$\boldsymbol{\Delta}(\rho, \boldsymbol{\beta})\mathbf{z}_{.t} = \boldsymbol{\Phi}\mathbf{f}_t + \mathbf{u}_{.t}, \text{ for } t = 1, 2, ..., T, \tag{3}$$

where $\mathbf{z}_{.t} = (\mathbf{z}_{1t}', \mathbf{z}_{2t}', ..., \mathbf{z}_{Nt}')'$ is an $N(k+1)$-dimensional vector of observations, with $\mathbf{z}_{it} = (y_{it}, \mathbf{x}_{it}')'$, $\boldsymbol{\Phi} = (\boldsymbol{\Phi}_1, \boldsymbol{\Phi}_2, ..., \boldsymbol{\Phi}_N)'$, $\mathbf{u}_{.t} = (\mathbf{u}_{1t}', \mathbf{u}_{2t}', ..., \mathbf{u}_{Nt}')'$, and $\boldsymbol{\Delta} = \boldsymbol{\Delta}(\rho, \boldsymbol{\beta})$ is a square matrix, of which the $(i,j)^{th}$ subblock of size $(k+1)$, for $i, j = 1, 2, ..., N$, is given by

$$\boldsymbol{\Delta}_{ii} = \begin{pmatrix} 1 & -\boldsymbol{\beta}' \\ \mathbf{0}_{k\times 1} & \mathbf{I}_k \end{pmatrix}, \text{ if } i = j; \text{ and } \boldsymbol{\Delta}_{ij} = \begin{pmatrix} -\rho w_{ij} & \mathbf{0}_{1\times k} \\ \mathbf{0}_{k\times 1} & \mathbf{0}_{k\times k} \end{pmatrix}, \text{ if } i \neq j.$$

The way of stacking the equations in (2) follows that in Bai and Li (2014), who show that $\boldsymbol{\Delta}^{-1} = \boldsymbol{\Delta}^{-1}(\rho, \boldsymbol{\beta})$ exists and its $(i,j)^{th}$ subblock is given by[8]

$$\boldsymbol{\Delta}_{ii}^{-1} = \begin{pmatrix} \check{s}_{ii} & \check{s}_{ii}\boldsymbol{\beta}' \\ \mathbf{0}_{k\times 1} & \mathbf{I}_k \end{pmatrix}, \text{ if } i = j; \text{ and } \boldsymbol{\Delta}_{ij}^{-1} = \begin{pmatrix} \check{s}_{ij} & \check{s}_{ij}\boldsymbol{\beta}' \\ \mathbf{0}_{k\times 1} & \mathbf{0}_{k\times k} \end{pmatrix}, \text{ if } i \neq j, \tag{4}$$

where $\check{s}_{ij}$ denotes the $(i,j)^{th}$ element of $\mathbf{S}^{-1}(\rho)$, and $\mathbf{S}(\rho) = \mathbf{I}_N - \rho\mathbf{W}$. The inverse of $\mathbf{S}(\rho)$ exists under certain regularity conditions, which will be discussed below. It then follows from (4) that (3) is equivalent to

$$\mathbf{z}_{.t} = \boldsymbol{\Delta}^{-1}(\boldsymbol{\Phi}\mathbf{f}_t + \mathbf{u}_{.t}) = \mathbf{C}'\mathbf{f}_t + \boldsymbol{\epsilon}_{.t}, \tag{5}$$

where $\mathbf{C} = (\boldsymbol{\Delta}^{-1}\boldsymbol{\Phi})'$ and $\boldsymbol{\epsilon}_{.t} = \boldsymbol{\Delta}^{-1}\mathbf{u}_{.t} = (\boldsymbol{\epsilon}_{1t}', \boldsymbol{\epsilon}_{2t}', ..., \boldsymbol{\epsilon}_{Nt}')'$ are the transformed new error terms.

Now letting $\boldsymbol{\Theta}_a = N^{-1}\boldsymbol{\tau}_N' \otimes \mathbf{I}_{k+1}$, where $\boldsymbol{\tau}_N$ is an $N \times 1$ vector of ones, it is easily verified that $\bar{\mathbf{z}}_{.t} = \boldsymbol{\Theta}_a\mathbf{z}_{.t} = (\bar{y}_{.t}, \bar{\mathbf{x}}_{.t}')'$, where $\bar{y}_{.t} = N^{-1}\sum_{i=1}^{N} y_{it}$ and $\bar{\mathbf{x}}_{.t} = N^{-1}\sum_{i=1}^{N}\mathbf{x}_{it}$. As shown, $\boldsymbol{\Theta}_a$ is a matrix that operates on any $N(k+1)$-dimensional vector that is stacked in the same order as $\mathbf{z}_{.t}$ and produces an $k \times 1$ vector of cross-sectional averages. Similarly, we have $\bar{\boldsymbol{\epsilon}}_{.t} = \boldsymbol{\Theta}_a\boldsymbol{\epsilon}_{.t} = N^{-1}\sum_{i=1}^{N}\boldsymbol{\epsilon}_{it}$. Premultiplying both sides of (5) with $\boldsymbol{\Theta}_a$ yields

$$\bar{\mathbf{z}}_{.t} = \bar{\mathbf{C}}'\mathbf{f}_t + \bar{\boldsymbol{\epsilon}}_{.t}, \tag{6}$$

where

$$\bar{\mathbf{C}} = (\boldsymbol{\Theta}_a\mathbf{C}')' = \left[\frac{1}{N}\sum_{j=1}^{N}\check{s}_{.j}(\gamma_j + \mathbf{A}_j\boldsymbol{\beta}), \frac{1}{N}\sum_{j=1}^{N}\mathbf{A}_j\right], \tag{7}$$

and $\check{s}_{.j} = \sum_{i=1}^{N}\check{s}_{ij}$. Note that the first column of $\bar{\mathbf{C}}$ comprises weighted averages of factor loadings, where the weights are column sums of the spatial matrix $\mathbf{S}^{-1}(\rho)$. In the absence of spatial effects $(\rho = 0)$, $\bar{\mathbf{C}}$ would reduce to the same matrix as defined in Pesaran (2006) consisting of only simple averages. Assuming that $\bar{\mathbf{C}}$ has full row rank, namely, Rank $(\bar{\mathbf{C}}) = m \leq k+1$, for all N including $N \to \infty$, we obtain

[8]See Lemma A.1 of Bai and Li (2014).

$$\mathbf{f}_t = \left(\bar{\mathbf{C}}\bar{\mathbf{C}}'\right)^{-1}\bar{\mathbf{C}}(\bar{\mathbf{z}}_{.t} - \bar{\boldsymbol{\epsilon}}_{.t}). \tag{8}$$

The task now is to show that $\bar{\boldsymbol{\epsilon}}_{.t}$ diminishes for sufficiently large N. We establish in Lemma S.2 that $\bar{\boldsymbol{\epsilon}}_{.t}$ converges to zero in quadratic mean as $N \to \infty$, for any t.[9] It then follows from (8) that $\mathbf{f}_t$ can be approximated by the cross-sectional averages $\bar{\mathbf{z}}_{.t}$ with an error of order $O_p\left(1/\sqrt{N}\right)$. More formally, we have

$$\mathbf{f}_t = \left(\mathbf{C}_0\mathbf{C}_0'\right)^{-1}\mathbf{C}_0\bar{\mathbf{z}}_{.t} + o_p(1), \text{ as } N \to \infty, \tag{9}$$

where $\mathbf{C}_0 = \lim_{N\to\infty}\bar{\mathbf{C}}$. It is clear from (9) that $\bar{\mathbf{z}}_{.t}$ can serve as factor proxies as long as N is large.[10] Note that the use of equal weights in constructing the cross-sectional averages is nonessential to the asymptotic analysis, which can be readily carried through with other weighting schemes satisfying the granularity conditions.[11] Thus, the current article will focus on simple cross-sectional averages for ease of exposition.

Note that the reduced form of (1) can be written as

$$\mathbf{y}_{.t} = \mathbf{S}^{-1}(\rho)(\mathbf{X}_{.t}\boldsymbol{\beta} + \boldsymbol{\Gamma}\mathbf{f}_t + \mathbf{e}_{.t}), \text{ for } t = 1, 2, ..., T,$$

where $\mathbf{y}_{.t} = (y_{1t}, y_{2t}, ..., y_{Nt})', \mathbf{X}_{.t} = (\mathbf{x}_{1t}, \mathbf{x}_{2t}, ..., \mathbf{x}_{Nt})'$, $\boldsymbol{\Gamma} = (\boldsymbol{\gamma}_1, \boldsymbol{\gamma}_2, ..., \boldsymbol{\gamma}_N)'$, $\mathbf{e}_{.t} = (e_{1t}, e_{2t}, ..., e_{Nt})'$. We use $\boldsymbol{\delta}_0 = (\rho_0, \boldsymbol{\beta}_0')'$ to denote the true parameter values of $\boldsymbol{\delta} = (\rho, \boldsymbol{\beta}')'$. The following matrices will be frequently used throughout the current article:

$$\begin{aligned} \mathbf{S}_0 &= \mathbf{S}(\rho_0) = \mathbf{I}_N - \rho_0\mathbf{W}, \\ \mathbf{G}(\rho) &= \mathbf{W}\mathbf{S}^{-1}(\rho), \mathbf{G}_0 = \mathbf{W}\mathbf{S}_0^{-1}, \mathbf{G}_0^b = \mathbf{I}_T \otimes \mathbf{G}_0. \end{aligned} \tag{10}$$

In addition, to facilitate analysis it is convenient to define the infeasible de-factoring matrices as follows:

$$\mathbf{M}_f = \mathbf{I}_T - \mathbf{F}(\mathbf{F}'\mathbf{F})^{-}\mathbf{F}', \quad \mathbf{M}_f^b = \mathbf{M}_f \otimes \mathbf{I}_N, \tag{11}$$

where $\mathbf{F} = (\mathbf{f}_1, \mathbf{f}_2, ..., \mathbf{f}_T)'$ is a $T \times m$ matrix of unobserved common factors, and $(\mathbf{F}'\mathbf{F})^{-}$ denotes the generalized inverse of $\mathbf{F}'\mathbf{F}$. The observable counterparts of (11) that utilize cross-sectional averages are given by

$$\bar{\mathbf{M}} = \mathbf{I}_T - \bar{\mathbf{Z}}(\bar{\mathbf{Z}}'\bar{\mathbf{Z}})^{-}\bar{\mathbf{Z}}', \ \mathbf{M}^b = \bar{\mathbf{M}} \otimes \mathbf{I}_N, \tag{12}$$

where $\bar{\mathbf{Z}} = (\bar{\mathbf{z}}_{.1}, \bar{\mathbf{z}}_{.2}, ..., \bar{\mathbf{z}}_{.T})'$. Note that $\mathbf{M}_f^b$ and $\mathbf{M}^b$ are de-factoring matrices of NT dimension that operate on the observations stacked as successive cross-sections, namely, $\mathbf{Y} = (\mathbf{y}_{.1}', \mathbf{y}_{.2}', ..., \mathbf{y}_{.T}')'$ and $\mathbf{X} = (\mathbf{X}_{.1}', \mathbf{X}_{.2}', ..., \mathbf{X}_{.T}')'$. It is also convenient to use $\mathbf{L} = (\mathbf{Y}^*, \mathbf{X})$ to denote the full set of regressors, where $\mathbf{L} = (\mathbf{L}_{.1}', \mathbf{L}_{.2}', ..., \mathbf{L}_{.T}')'$ is of dimension $NT \times (k+1)$, with $\mathbf{L}_{.t} = (\boldsymbol{l}_{1t}, \boldsymbol{l}_{2t}, ..., \boldsymbol{l}_{Nt})'$ and $\boldsymbol{l}_{it} = (y_{it}^*, \mathbf{x}_{it}')'$. Throughout this article, K is used generically to denote a finite positive constant. For any matrix $\mathbf{A}$, $||\mathbf{A}||$, $||\mathbf{A}||_\infty$, and $||\mathbf{A}||_1$ denote the Frobenius norm, maximum row sum norm, and maximum column sum norm, respectively. See Appendix A for a summary of frequently used notations.

In order to formally analyze model (1), we make the following assumptions.

Assumption 1. The unobserved common factors $\mathbf{f}_t$ are covariance stationary with absolutely summable autocovariances, and they are distributed independently of $e_{it'}$ and $\mathbf{v}_{it'}$ for all i, t, t'.

Assumption 2. The idiosyncratic errors, e_{it} and $\mathbf{v}_{it}$, follow linear stationary processes with absolutely summable autocovariances for each i: $e_{it} = \sum_{l=0}^{\infty} a_{il}\zeta_{i,t-l}$ and $\mathbf{v}_{it} = \sum_{l=0}^{\infty} \boldsymbol{\Xi}_{il}\boldsymbol{\varsigma}_{i,t-l}$, where $(\zeta_{it}, \boldsymbol{\varsigma}_{it}')' \sim IID(\mathbf{0}_{k+1}, \mathbf{I}_{k+1})$ with finite fourth-order moments. $Var(e_{it}) = \sum_{l=0}^{\infty} a_{il}^2 = \sigma_i^2 < K$ and

[9] All lemmas are provided in the Supporting Information.
[10] In practice, it may also worth including $\bar{y}_t^* = N^{-1}\sum_{i=1}^N y_t^*$ as factor proxies if $\bar{y}_t^*$ is not highly correlated with $\bar{y}_t$.
[11] See Assumption 5 in Pesaran (2006).

$Var(\mathbf{v}_{it}) = \sum_{l=0}^{\infty} \mathbf{\Xi}_{il}\mathbf{\Xi}_{il}' = \mathbf{\Sigma}_{v,i} < K$, where $\sigma_i^2 > 0$ and $\mathbf{\Sigma}_{v,i}$ is positive definite. In addition, e_{it} and $\mathbf{v}_{jt'}$ are distributed independently of each other, for all i, j, t, t'.

Assumption 3. The factor loadings, $\boldsymbol{\gamma}_i$ and $\mathbf{A}_i$, are nonrandom such that $||\boldsymbol{\gamma}_i|| < K$ and $||\mathbf{A}_i|| < K$, for all i.

Assumption 4. The true parameter vector, $\boldsymbol{\delta}_0 = (\rho_0, \boldsymbol{\beta}_0')'$, is in the interior of the parameter space, denoted by Δ_{sp}, which is a compact subset of the $(k+1)$-dimensional Euclidean space, $\mathbb{R}^{k+1}$.

Assumption 5. The matrix $\bar{\mathbf{C}}$, given by (7), has full row rank for all N, including $N \to \infty$.

Assumption 6. The $N \times N$ nonstochastic spatial weights matrix, $\mathbf{W} = (w_{ij})$, has bounded maximum row and column sum norms, namely, $||\mathbf{W}||_\infty < K$ and $||\mathbf{W}||_1 < K$, respectively, and $|\rho| < \max\{1/||\mathbf{W}||_1, 1/||\mathbf{W}||_\infty\}$ for all values of ρ in the parameter space. The diagonal entries of $\mathbf{W}$ are zero, i.e., $w_{ii} = 0$, for all i.

Assumption 7. The $N \times$ q matrix of instrumental variables, $\mathbf{Q}_{.t}$, for $t = 1, 2, ..., T$, is composed of a subset of the columns of $(\mathbf{X}_{.t}, \mathbf{W}\mathbf{X}_{.t}, \mathbf{W}^2\mathbf{X}_{.t}, ..., \mathbf{W}^R\mathbf{X}_{.t})$, and its column dimension q is fixed for all N and t. The matrix $\mathbf{Q} = (\mathbf{Q}_{.1}', \mathbf{Q}_{.2}', ..., \mathbf{Q}_{.T}')'$ represents the IV matrix of dimension $NT \times$ q.

Assumption 8.

(a) There exist N_0 and T_0, such that for all $N > N_0$ and $T > T_0$, the matrices $(NT)^{-1}\mathbf{Q}'\mathbf{M}^b\mathbf{Q}$ and $(NT)^{-1}\mathbf{Q}'\mathbf{M}_f^b\mathbf{Q}$ exist and are nonsingular, and $(NT)^{-1}\mathbf{Q}'\mathbf{M}_f^b\mathbf{L}_0$ is of full column rank, where

$$\mathbf{L}_0 = \left(\mathbf{G}_0^b\mathbf{X}\boldsymbol{\beta}_0, \mathbf{X}\right), \quad \mathbf{G}_0^b = \mathbf{I}_T \otimes \mathbf{G}_0, \tag{13}$$

and $\mathbf{G}_0$ is given by (10).

(b) $\sup_{i,t,p} E|x_{it,p}|^{2+\delta} < K$, for some $\delta > 0$, and for $i = 1, 2, ..., N, t = 1, 2, ..., T$, and $p = 1, 2, ..., k$.

Remark 1. The stationarity assumption for the factors in Assumption 1 is standard in the CCE literature.[12] Assumption 2 for the error processes is quite general as it allows for the presence of both unknown heteroskedasticity and serial correlation.[13] Note that Assumption 2 implies that $Var(\mathbf{u}_{.t}) = \mathbf{\Sigma}_u = Diag(\mathbf{\Sigma}_{u,1}, \mathbf{\Sigma}_{u,2}, ..., \mathbf{\Sigma}_{u,N})$ and $\mathrm{Var}(\mathbf{u}_{it}) = \mathbf{\Sigma}_{u,i} = Diag(\sigma_i^2, \mathbf{\Sigma}_{v,i})$, for $i = 1, 2, ..., N$; both $\mathbf{\Sigma}_u$ and $\mathbf{\Sigma}_{u,i}$ are block-diagonal matrices.

Remark 2. The nonrandom factor loadings imposed by Assumption 3 are more general than random loadings and imply that $\mathbf{x}_{it}$ can be endogenous through unobserved factors. The compactness of the parameter space in Assumption 4 is a condition to facilitate theoretical analysis of the GMM estimation. This condition is usually assumed when the objective function for an estimator is highly nonlinear. The rank condition in Assumption 5 is imposed for analytical convenience and can be relaxed following similar arguments as in Pesaran (2006).[14]

Remark 3. The assumption that $\mathbf{W}$ has uniformly bounded row and column norms in Assumption 6 is standard in the spatial econometrics literature. It essentially imposes sparsity

[12] An exception is made by Kapetanios, Pesaran, and Yamagata (2011), who show that the CCE approach can be extended to allow latent factors to follow unit root processes.

[13] This model can be further extended to accommodate spatial correlations in the errors at the expense of more involved notations.

[14] See Karabiyik, Reese, and Westerlund (2017) for a recent investigation on the role of the rank condition in CCE estimations.

restrictions on $\mathbf{W}$ so that the degree of cross-sectional correlation is tractable. As we shall see, this assumption plays an important role in the asymptotic analysis.[15] Also note that $\mathbf{W}$ need not be row-standardized such that each row sums to unity, which is often performed in practice to aid interpretation. If all the elements of $\mathbf{W}$ are non-negative, row-standardization implies that y_{it}^{*} is a weighted average of neighboring values. The zero-diagonal assumption for the $\mathbf{W}$ matrix is innocuous and only for notational convenience in discussing the GMM estimation. No unit has self-influence under this assumption, which is satisfied if $\mathbf{W}$ represents geographical distance or social interactions. Last but not least, Assumption 6 ensures that $\mathbf{S}(\rho) = \mathbf{I}_N - \rho\mathbf{W}$ is nonsingular for all possible values of ρ. To see this, note that $\mathbf{S}(\rho)$ is invertible if $|\lambda_{\max}(\rho\mathbf{W})| < 1$. Since $\lambda_{\max}(\rho\mathbf{W}) < |\rho|||\mathbf{W}||_1$ and $\lambda_{\max}(\rho\mathbf{W}) < |\rho|||\mathbf{W}||_\infty$, therefore $\mathbf{S}(\rho)$ is invertible if $|\rho| < \max\{1/||\mathbf{W}||_1, 1/||\mathbf{W}||_\infty\}$. Assumption 6 also implies that $\mathbf{S}^{-1}(\rho)$ is uniformly bounded in row and column sums in absolute value for all values of ρ, since

$$||\mathbf{S}^{-1}(\rho)||_1 = ||\mathbf{I}_N + \rho\mathbf{W} + \rho^2\mathbf{W}^2 + ...||_1 \leq 1 + |\rho|||\mathbf{W}||_1 + |\rho|^2||\mathbf{W}||_1^2 + ... = \frac{1}{1 - |\rho|||\mathbf{W}||_1} < K,$$

and similarly, it can be shown that $||\mathbf{S}^{-1}(\rho)||_\infty < K$. The boundedness of $||\mathbf{S}^{-1}(\rho)||_1$ together with the uniform boundedness of the factor loadings (Assumption 3) imply that the elements of $\bar{\mathbf{C}}$ are bounded uniformly.

Remark 4. The spatially lagged dependent variable, y_{it}^{*}, is generally correlated with the error term and results in endogeneity. The selection of the instrumental variables in Assumption 7 originates from Kelejian and Prucha (1998) for cross-section SAR models. This choice is motivated by the spatial power series expansion of the expectation of the spatially lagged dependent variable (Kelejian and Prucha, 1998, p. 104).[16] Also note that Assumption 6 implies that $\mathbf{W}^r$ also has bounded row and column sum norms, for $r = 1, 2, ..., R$.

Remark 5. Assumption 8(a) ensures that true parameter vector, $\boldsymbol{\delta}_0$, is identified based on the IV/GMM framework. Detailed explanations will be given in the next section. The existence of higher-than-second moments in Assumption 8(b) is required for the GMM estimation to apply a central limit theorem (CLT) for the linear and quadratic form, which is an extension of Theorem 1 in Kelejian and Prucha (2001). For the 2SLS estimations, the existence of the second moments would be sufficient.

3. Estimation

In this section, we suggest three estimation methods for model (1), including the 2SLS, Best 2SLS, and GMM estimations, and formally establish the asymptotic distributions of these estimators.

3.1. 2SLS estimation

The first estimation method we propose is the 2SLS estimation using the instrumental variables, $\mathbf{Q}$, as specified in Assumption 7. As before, $\boldsymbol{\delta}_0 = (\rho_0, \boldsymbol{\beta}_0')'$ denotes the true parameter vector.[17] The 2SLS estimator of $\boldsymbol{\delta}_0$, denoted by $\hat{\boldsymbol{\delta}}_{2sls}$, is defined as

[15] Pesaran and Yang (2020b) made a recent effort to relax this assumption for cross-section SAR models.

[16] The highest order of the power of $\mathbf{W}$, denoted by R, is a preselected finite constant, and in practice is usually chosen to be at least two.

[17] We do not use subscript "0" to refer to the true values of γ_i, $\mathbf{A}_i, \sigma_i^2$, and $\boldsymbol{\Sigma}_{v,i}$ in the main text in order to avoid notational complications.

$$\hat{\boldsymbol{\delta}}_{2sls} = \left(\mathbf{L}'\mathbf{P}_Q\mathbf{L}\right)^{-1}\mathbf{L}'\mathbf{P}_Q\mathbf{Y}, \tag{14}$$

where $\mathbf{P}_Q = \mathbf{M}^b\mathbf{Q}(\mathbf{Q}'\mathbf{M}^b\mathbf{Q})^{-1}\mathbf{Q}'\mathbf{M}^b$, $\mathbf{L} = (\mathbf{Y}^*, \mathbf{X})$ and $\mathbf{M}^b$ is given by (12). There are two ways to interpret (14). One way is to de-factor the data with cross-sectional averages, namely, $\overset{\circ}{\mathbf{Y}} = \mathbf{M}^b\mathbf{Y}$ and $\overset{\circ}{\mathbf{L}} = \mathbf{M}^b\mathbf{L}$, and then apply the standard 2SLS procedure to the de-factored observations $\overset{\circ}{\mathbf{Y}}$ and $\overset{\circ}{\mathbf{L}}$. Alternatively, the matrix $\mathbf{M}^b\mathbf{Q}$ can be directly viewed as instruments.

The following theorem summarizes the asymptotic properties of the 2SLS estimator. The proof is given in Appendix B.

Theorem 1. *Consider the panel data model given by (1) and suppose that Assumptions 1–8 hold. Then,*

(a) *the 2SLS estimator of $\boldsymbol{\delta}_0$, denoted by $\hat{\boldsymbol{\delta}}_{2sls}$ and defined in (14), is consistent for $\boldsymbol{\delta}_0$, as $N \to \infty$, for T fixed or $T \to \infty$,*
(b) *as $(N,T) \xrightarrow{j} \infty$ with $\sqrt{T}/N \to 0$, we have*

$$\sqrt{NT}\left(\hat{\boldsymbol{\delta}}_{2sls} - \boldsymbol{\delta}_0\right) \sim \boldsymbol{\Psi}_{LPL}^{-1}\boldsymbol{\Psi}'_{QML}\boldsymbol{\Psi}_{QMQ}^{-1}\sqrt{\frac{T}{N}}\boldsymbol{b}_{2sls} + N(\mathbf{0}, \boldsymbol{\Sigma}_{2sls}), \tag{15}$$

where

$$\boldsymbol{\Sigma}_{2sls} = \boldsymbol{\Psi}_{LPL}^{-1}\boldsymbol{\Omega}_{LPe}\boldsymbol{\Psi}_{LPL}^{-1}, \tag{16}$$

$$\boldsymbol{\Psi}_{LPL} = \operatorname*{plim}_{N,T\to\infty}\frac{1}{NT}\mathbf{L}_0'\mathbf{P}_{Q,f}\mathbf{L}_0, \quad \boldsymbol{\Omega}_{LPe} = \boldsymbol{\Psi}'_{QML}\boldsymbol{\Psi}_{QMQ}^{-1}\boldsymbol{\Omega}_{QMe}\boldsymbol{\Psi}_{QMQ}^{-1}\boldsymbol{\Psi}_{QML}, \tag{17}$$

$$\boldsymbol{\Psi}_{QMQ} = \operatorname*{plim}_{N,T\to\infty}\frac{1}{NT}\mathbf{Q}'\mathbf{M}_f^b\mathbf{Q}, \quad \boldsymbol{\Psi}_{QML} = \operatorname*{plim}_{N,T\to\infty}\frac{1}{NT}\mathbf{Q}'\mathbf{M}_f^b\mathbf{L}_0, \tag{18}$$

$$\boldsymbol{\Omega}_{QMe} = \lim_{N\to\infty}\left(\frac{1}{N}\sum_{i=1}^{N}\boldsymbol{\Omega}_{iQMe}\right), \quad \boldsymbol{\Omega}_{iQMe} = \operatorname*{plim}_{T\to\infty}\frac{1}{T}\mathbf{Q}_{i.}'\mathbf{M}_f\boldsymbol{\Omega}_{e,i}\mathbf{M}_f\mathbf{Q}_{i.}, \tag{19}$$

$$\mathbf{P}_{Q,f} = \mathbf{M}_f^b\mathbf{Q}\left(\mathbf{Q}'\mathbf{M}_f^b\mathbf{Q}\right)^{-1}\mathbf{Q}'\mathbf{M}_f^b, \quad \boldsymbol{\Omega}_{e,i} = E\left(\mathbf{e}_{i.}\mathbf{e}_{i.}'\right), \tag{20}$$

$$\begin{gathered}\mathbf{Q}_{i.} = (\mathbf{Q}_{i1}, \mathbf{Q}_{i2}, ..., \mathbf{Q}_{iT})', \quad \mathbf{e}_{i.} = (e_{i1}, e_{i2}, ..., e_{iT})' \text{ and} \\ \boldsymbol{b}_{2sls} = \boldsymbol{b}_1 - \boldsymbol{b}_2 - \boldsymbol{b}_3,\end{gathered} \tag{21}$$

where the elements of $\boldsymbol{b}_1, \boldsymbol{b}_2$, and $\boldsymbol{b}_3$ resulting from a particular choice of r and p, for $r = 1, 2, ..., R, p = 1, 2, ..., k$, are given by

$$b_{1,rp} = \lim_{N\to\infty}\frac{1}{N}\sum_{i=1}^{N}\sum_{l=1}^{N}w_{il}^{(r)}\boldsymbol{s}_p'\mathbf{A}_l'(\bar{\mathbf{C}}^-)'\bar{\boldsymbol{\Sigma}}_\epsilon\bar{\mathbf{C}}^-\boldsymbol{\gamma}_i, \tag{22}$$

$$b_{2,rp} = \lim_{N\to\infty}\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{N}\sum_{l=1}^{N}w_{il}^{(r)}\boldsymbol{s}_p'(\mathbf{0}_{k\times 1}, \boldsymbol{\Sigma}_{v,l})\boldsymbol{\Delta}_{jl}^{-1'}\bar{\mathbf{C}}^-\boldsymbol{\gamma}_i, \tag{23}$$

$$b_{3,rp} = \lim_{N\to\infty}\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{N}\sum_{l=1}^{N}w_{il}^{(r)}\boldsymbol{s}_p'\mathbf{A}_l'(\bar{\mathbf{C}}^-)'\boldsymbol{\Delta}_{ji}^{-1}\left(\sigma_i^2, \mathbf{0}_{1\times k}\right)', \tag{24}$$

$$\bar{\boldsymbol{\Sigma}}_\epsilon = \frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{N}\sum_{l=1}^{N}\boldsymbol{\Delta}_{il}^{-1}\boldsymbol{\Sigma}_{u,l}\boldsymbol{\Delta}_{jl}^{-1'}, \quad \boldsymbol{\Sigma}_{u,i} = E\left(\mathbf{u}_{it}\mathbf{u}_{it}'\right) = Diag\left(\sigma_i^2, \boldsymbol{\Sigma}_{v,i}\right), \tag{25}$$

respectively, $w_{ij}^{(r)}$ *denotes the* $(i,j)^{th}$ *element of* $\mathbf{W}^r, \mathbf{W}^0 \equiv \mathbf{I}_N$, $\boldsymbol{s}_p$ *is the* $k \times 1$ *selection vector with one as its pth element and zeros elsewhere,* $\bar{\mathbf{C}}^- = \bar{\mathbf{C}}'(\bar{\mathbf{C}}\bar{\mathbf{C}}')^{-1}$, *and* $\boldsymbol{\Delta}_{ji}^{-1}$ *is defined in (4).*

First, it is worth noting that the consistency of the 2SLS estimator does not require T to be large. Part (a) of Theorem 1 shows that as long as N is large, the 2SLS estimator will be consistent irrespective of the size of T. This is in congruent with the result in (9), which suggests that the unknown factors can be approximated by cross-sectional averages with an error of order $O_p\left(1/\sqrt{N}\right)$. Second, part (b) of Theorem 1 states that the 2SLS estimator will be normally distributed if N and T tend to infinity jointly with $\sqrt{T}/N \to 0$, which is satisfied if $T < N^2$. Specifically, the asymptotic distribution of the 2SLS estimator consists of two terms: a bias term involving $\sqrt{T/N}\boldsymbol{b}_{2sls}$ and a normal variate with mean zero and variance-covariance matrix $\boldsymbol{\Sigma}_{2sls}$. Some remarks about the asymptotic bias term are in order.

Remark 6. Clearly, if $T/N \to 0$ as $(N,T) \xrightarrow{j} \infty$, the bias term will vanish and then the distribution of $\hat{\boldsymbol{\delta}}_{2sls}$ will become $\sqrt{NT}(\hat{\boldsymbol{\delta}}_{2sls} - \boldsymbol{\delta}_0) \sim N(\mathbf{0}, \boldsymbol{\Sigma}_{2sls})$. That is, the asymptotic distribution no longer involves nuance parameters. This result is consistent with the finding for the pooled CCE (CCEP) estimator in the case of homogeneous slopes in Pesaran (2006), and also the results in Westerlund and Urbain (2015) and Karabiyik, Reese, and Westerlund (2017). Also note that this is the same distribution as if the factors were known. In other words, there will be no asymptotic bias if factors are observable.

Remark 7. If the factors are unobservable and $T/N \to \kappa > 0$ as $(N,T) \xrightarrow{j} \infty$, where κ is a finite positive constant, then there exists the bias term that comprises three components: $\boldsymbol{b}_1$, $\boldsymbol{b}_2$, and $\boldsymbol{b}_3$. First, $\boldsymbol{b}_1$ arises from the uncertainty in estimating the factors from an OLS regression of $\bar{\mathbf{z}}_{.t}$ on $\bar{\mathbf{C}}$ following (8) as if $\bar{\mathbf{C}}$ were known; the term, $(\bar{\mathbf{C}}^-)'\bar{\boldsymbol{\Sigma}}_\epsilon\bar{\mathbf{C}}^-$, in $\boldsymbol{b}_1$ is the covariance matrix of this OLS estimator. $\boldsymbol{b}_2$ and $\boldsymbol{b}_3$ exist due to the correlations between the cross-sectional averages and the idiosyncratic errors, $\mathbf{v}_{it}$ and $e_{it.}$, respectively; these correlations are evident from the terms $\boldsymbol{\Sigma}_{v,l}$ and σ_i^2. The elements of $\mathbf{W}^r$ show up in the bias term owning to the usage of $\mathbf{W}^r\mathbf{X}_{.t}$ in the IV matrix.

Remark 8. The result implied by Theorem 1 when T/N tends to a finite positive constant is in agreement with the findings in Westerlund and Urbain (2015), where the authors consider panel data models with latent factors but no spatial dependence and show that the $\sqrt{NT}$-consistent CCE estimator is asymptotically biased when $T/N \to \kappa > 0$.[18]

The next remark concerns the identification of $\boldsymbol{\delta}_0$.

Remark 9. The full rank condition in Assumption 8(a) ensures the identifiability of $\boldsymbol{\delta}_0$. To see this, $\boldsymbol{\delta}_0$ is identified if $\lim_{N\to\infty} E\left[(NT)^{-1}\mathbf{L}'\mathbf{P}_Q\mathbf{L}\right]$ is nonsingular, which holds if and only if $\lim_{N\to\infty} E\left[(NT)^{-1}\mathbf{Q}'\mathbf{M}^b\mathbf{L}\right]$ has full column rank. Since $(NT)^{-1}\mathbf{Q}'\mathbf{M}^b\mathbf{L}$ converges to $(NT)^{-1}\mathbf{Q}'\mathbf{M}_f^b\mathbf{L}_0$ in probability as $N \to \infty$ (as shown in (B.4) of Appendix B), identification is obtained if $(NT)^{-1}\mathbf{Q}'\mathbf{M}_f^b\mathbf{L}_0$ has full rank. It is worth noting that the identification condition is the same whether the factors are observable or not. If factors are unobservable, they can be identified (up to a rotation) by CCE or principal component approaches, without knowing ρ and $\mathbf{W}$. Note that Eq. (6) is a factor model with weakly cross-sectionally dependent errors and hence the factors can be estimated by (8). Thereby the unknown factors are well approximated by cross-sectional averages for all values of ρ and $\boldsymbol{\beta}$ under the given assumptions with an approximation error of order

[18]Karabiyik, Reese, and Westerlund (2017) recently point out a common problem with the CCE approach and show that in the degenerate case when $m < k+1$ and when T/N tends to a finite positive constant as $(N,T) \xrightarrow{j} \infty$, there will be an additional bias term. They assume independently and identically distributed errors and only focus on $T/N = O(1)$. Similar additional bias term is expected to occur for our model under these conditions. A more general extension of Theorem 1 and the investigation of bias correction for the case of $T/N = O(1)$ is left for future work.

$O_p\left(1/\sqrt{N}\right)$. Hence, replacing the unobserved factors in model (1) with their consistent estimates leads to the same identification condition as when factors are observable.

A consistent estimator for the asymptotic variance matrix, $\boldsymbol{\Sigma}_{2sls}$, is given by[19]

$$\hat{\boldsymbol{\Sigma}}_{2sls} = \hat{\boldsymbol{\Psi}}_{LPL}^{-1}\hat{\boldsymbol{\Omega}}_{LPe}\hat{\boldsymbol{\Psi}}_{LPL}^{-1}, \tag{26}$$

where $\hat{\boldsymbol{\Psi}}_{LPL} = (NT)^{-1}\mathbf{LP}_Q\mathbf{L}$, and $\hat{\boldsymbol{\Omega}}_{LPe}$ can be obtained by a Newey and West (1987) type robust estimator following the suggestion by Pesaran (2006):

$$\begin{aligned}
\hat{\boldsymbol{\Omega}}_{LPe} &= N^{-1}\sum_{i=1}^{N}\hat{\boldsymbol{\Omega}}_{iLPe}, \\
\hat{\boldsymbol{\Omega}}_{iLPe} &= \hat{\boldsymbol{\Omega}}_{iLPe,0} + \sum_{h=1}^{M_h}\left(1-\frac{h}{M_h+1}\right)\left(\hat{\boldsymbol{\Omega}}_{iLPe,h} + \hat{\boldsymbol{\Omega}}'_{iLPe,h}\right), \\
\hat{\boldsymbol{\Omega}}_{iLPe,h} &= T^{-1}\sum_{t=h+1}^{T}\hat{e}_{it}\hat{e}_{i,t-h}\hat{\boldsymbol{l}}_{it}\hat{\boldsymbol{l}}'_{i,t-h},
\end{aligned} \tag{27}$$

where M_h is the the window size of the Bartlett kernel, $\hat{\mathbf{e}} = \mathbf{M}^b(\mathbf{Y} - \mathbf{L}\hat{\boldsymbol{\delta}}_{2sls}) = (\hat{\mathbf{e}}'_{.1}, \hat{\mathbf{e}}'_{.2}, ..., \hat{\mathbf{e}}'_{.T})'$, $\hat{e}_{it}$ is the tth element of $\hat{\mathbf{e}}_{.t}$, $\hat{\mathbf{L}} = \mathbf{P}_Q\mathbf{L} = (\hat{\mathbf{L}}'_{.1}, \hat{\mathbf{L}}'_{.2}, ..., \hat{\mathbf{L}}'_{.T})'$, and $\hat{\boldsymbol{l}}'_{it}$ is the ith row of $\hat{\mathbf{L}}_{.t}$.

If $T/N \to \kappa > 0$ as $(N,T) \underset{j}{\to} \infty$, a bias-corrected 2SLS estimator may be proposed following Westerlund and Urbain (2015): $\hat{\boldsymbol{\delta}}_{2sls,bc} = \hat{\boldsymbol{\delta}}_{2sls} - \hat{\boldsymbol{\Psi}}_{LPL}^{-1}\hat{\boldsymbol{\Psi}}'_{QML}\hat{\boldsymbol{\Psi}}_{QMQ}^{-1}N^{-1}\hat{\boldsymbol{b}}_{2sls}$, where the estimated bias term $\hat{\boldsymbol{b}}_{2sls}$ is obtained as follows: $\boldsymbol{\Delta} = \boldsymbol{\Delta}(\boldsymbol{\delta}_0)$ is replaced with $\hat{\boldsymbol{\Delta}} = \boldsymbol{\Delta}(\hat{\boldsymbol{\delta}}_{2sls})$. The factor loadings $\boldsymbol{\gamma}_i$ can be estimated by regressing $y_{it} - \boldsymbol{l}'_{it}\hat{\boldsymbol{\delta}}_{2sls}$ on $\bar{\mathbf{z}}_{.t}$ with an intercept for each cross-section unit i; similarly, $\mathbf{A}_i$ can be estimated by regressing $\mathbf{x}_{it}$ on $\bar{\mathbf{z}}_{.t}$.[20] Then, $\boldsymbol{\gamma}_i$ and $\mathbf{A}_i$ are replaced with $\hat{\boldsymbol{\gamma}}_i$ and $\hat{\mathbf{A}}_i$, and the estimator $\hat{\bar{\mathbf{C}}}$ of $\bar{\mathbf{C}}$ also readily follows from $\hat{\boldsymbol{\gamma}}_i, \hat{\mathbf{A}}_i$ and the elements of $\mathbf{S}^{-1}(\hat{\rho}_{2sls})$. Finally, σ_i^2 is substituted by $\hat{\sigma}_i^2 = T^{-1}\sum_{t=1}^{T}\hat{e}_{it}^2$, and $\boldsymbol{\Sigma}_{v,i}$ is substituted by $\hat{\boldsymbol{\Sigma}}_{v,i} = T^{-1}\mathbf{X}'_i\bar{\mathbf{M}}\mathbf{X}_i$. We leave detailed investigation of bias correction under $T/N \underset{j}{\to} \kappa > 0$ for future studies; the current article will mainly focus on the case of $T/N \to 0$ as $(N,T) \underset{j}{\to} \infty$, which is relevant to our housing application and also suitable for many other data sets in practice.[21]

3.2. Best 2SLS estimation

Having established the asymptotic distribution of the 2SLS estimator, the question then naturally arises whether optimal instrumental variables are available for model (1). An instrument is considered optimal or "best" if it produces an estimator that has the smallest asymptotic variance among all the IV estimators for the model. For cross-sectional spatial models, Lee (2003) suggests a best generalized spatial 2SLS estimator and shows that it is asymptotically optimal under a set of regularity conditions. In this section, we generalize this estimation procedure to spatial models with common factors. Specifically, let $\hat{\boldsymbol{\delta}} = (\hat{\rho}, \hat{\boldsymbol{\beta}})$ denote some consistent initial estimate of $\boldsymbol{\delta}_0$, possibly obtained by the 2SLS estimation described in the previous section. We will investigate if the IV estimator, $\hat{\boldsymbol{\delta}}_{b2sls}$, can achieve the smallest asymptotic variance for model (1), where

$$\hat{\boldsymbol{\delta}}_{b2sls} = \left(\hat{\mathbf{Q}}^{*\prime}\mathbf{M}^b\mathbf{L}\right)^{-1}\hat{\mathbf{Q}}^{*\prime}\mathbf{M}^b\mathbf{Y}, \tag{28}$$

[19]See the Supporting Information for a proof that $\hat{\boldsymbol{\Sigma}}_{2sls}$ is consistent for $\boldsymbol{\Sigma}_{2sls}$.

[20]Although our interest lies in the parameters $\boldsymbol{\delta}$, we can gain insight into the variability of the factor loadings by running these regressions.

[21]The bias correction approach suggested by Westerlund and Urbain (2015) need to be reexamined in light of the latest contribution by Karabiyik, Reese, and Westerlund (2017), who show that there could be an additional bias term when $T/N = O(1)$ and $m < k+1$ but do not investigate bias correction.

$$\hat{\mathbf{Q}}^{*} = \left(\hat{\mathbf{G}}^{b}\mathbf{X}\hat{\boldsymbol{\beta}}, \mathbf{X}\right), \hat{\mathbf{G}}^{b} = \mathbf{I}_T \otimes \mathbf{G}(\hat{\rho}), \tag{29}$$

and as before $\mathbf{L} = (\mathbf{Y}^{*}, \mathbf{X})$ and $\mathbf{M}^{b}$ is the de-factoring matrix given by (12). We refer to $\hat{\boldsymbol{\delta}}_{b2sls}$ as the best 2SLS (B2SLS) estimator and $\hat{\mathbf{Q}}^{*}$ as the (feasible) best IV. Note that $\mathbf{L}_0$, which is defined by (13) and involves unknown true parameters ρ_0 and $\boldsymbol{\beta}_0$, corresponds to the infeasible best IV.

The intuition behind the formulation of $\hat{\mathbf{Q}}^{*}$ is to exploit the part of $\mathbf{Y}^{*}$ that is uncorrelated with the errors. To see this, suppose for simplicity that there are no common factors. The reduced form equation can be rewritten as: $\mathbf{y}_{.t} = \mathbf{S}_0^{-1}\mathbf{X}_{.t}\boldsymbol{\beta}_0 + \mathbf{S}_0^{-1}\mathbf{e}_{.t}$, which further leads to $\mathbf{y}^{*}_{.t} = \mathbf{G}_0\mathbf{X}_{.t}\boldsymbol{\beta}_0 + \mathbf{G}_0\mathbf{e}_{.t}$. It is readily seen that the term $\mathbf{G}_0\mathbf{X}_{.t}\boldsymbol{\beta}$ is correlated with $\mathbf{y}^{*}_{.t}$ but uncorrelated with $\mathbf{e}_{.t}$ given that $\mathbf{X}_{.t}$ is exogenous. Since $\mathbf{G}_0\mathbf{X}_{.t}\boldsymbol{\beta}_0$ depends on the unknown parameters ρ_0 and $\boldsymbol{\beta}_0$, a feasible IV for $\mathbf{y}^{*}_{.t}$ can be constructed as $\mathbf{G}(\hat{\rho})\mathbf{X}_{.t}\hat{\boldsymbol{\beta}}$. Accordingly, the B2SLS estimation can be implemented in two steps: first obtaining some preliminary consistent estimates of the parameters, and then conducting an IV estimation using the best IV based on the first-step estimates. A similar argument applies to model (1) containing common factors, with the modification that the common effects are filtered out using the de-factoring matrix $\mathbf{M}^{b}$.

The following theorem states the asymptotic properties of the B2SLS estimator and shows that it is the best IV estimator if T is much smaller than N and the error terms are independently and identically distributed. The proof is given in Appendix B.

Theorem 2. *Consider the panel data model given by (1). Suppose that Assumptions 1–6 and 8(b) hold, and* $(NT)^{-1}\mathbf{L}_0'\mathbf{M}_f^{b}\mathbf{L}_0$ *tends to a positive definite matrix as* $N \to \infty$ *for all* T *(including* $T \to \infty$*). Then,*

(a) *the best 2SLS (B2SLS) estimator,* $\hat{\boldsymbol{\delta}}_{b2sls}$*, defined in (28), is consistent for* $\boldsymbol{\delta}_0$*, as* $N \to \infty$*, for* T *fixed or* $T \underset{j}{\to} \infty$*,*

(b) *as* $(N, T) \underset{j}{\to} \infty$ *with* $\sqrt{T}/N \to 0$*, we have*

$$\sqrt{NT}\left(\hat{\boldsymbol{\delta}}_{b2sls} - \boldsymbol{\delta}_0\right) \sim \boldsymbol{\Psi}_{LML}^{-1}\sqrt{\frac{T}{N}}\boldsymbol{b}_{b2sls} + N(\mathbf{0}, \boldsymbol{\Sigma}_{b2sls}), \tag{30}$$

where

$$\begin{aligned} &\boldsymbol{\Sigma}_{b2sls} = \boldsymbol{\Psi}_{LML}^{-1}\boldsymbol{\Omega}_{LMe}\boldsymbol{\Psi}_{LML}^{-1}, \ \ \boldsymbol{\Psi}_{LML} = \underset{N,T\to\infty}{plim}\frac{1}{NT}\mathbf{L}_0'\mathbf{M}_f^{b}\mathbf{L}_0, \\ &\boldsymbol{\Omega}_{LMe} = \lim_{N\to\infty}\left(\frac{1}{N}\sum_{i=1}^{N}\boldsymbol{\Omega}_{iLMe}\right), \ \ \boldsymbol{\Omega}_{iLMe} = \underset{T\to\infty}{plim}\frac{1}{T}\mathbf{L}_{0,i.}'\mathbf{M}_f\boldsymbol{\Omega}_{e,i}\mathbf{M}_f\mathbf{L}_{0,i.} \end{aligned} \tag{31}$$

$\mathbf{L}_0$ *is given by (13),* $\mathbf{L}_{0,i.} = (\boldsymbol{l}_{0,i1}, \boldsymbol{l}_{0,i2}, ..., \boldsymbol{l}_{0,iT})'$*, and* $\boldsymbol{l}_{0,it}'$ *is the* $[N(t-1)+i]^{th}$ *row of* $\mathbf{L}_0, \boldsymbol{\Omega}_{e,i}$ *is given by (20),*

$$\begin{aligned} \boldsymbol{b}_{b2sls} &= \boldsymbol{b}_{1,b2sls} - \boldsymbol{b}_{2,b2sls} - \boldsymbol{b}_{3,b2sls}, \\ \boldsymbol{b}_{1,b2sls} &= \lim_{N\to\infty}\left[\frac{1}{N}\sum_{i=1}^{N}\sum_{l=1}^{N}g_{il}\boldsymbol{\beta}_0'\mathbf{A}_l'(\bar{\mathbf{C}}^{-})'\bar{\boldsymbol{\Sigma}}_{\epsilon}\bar{\mathbf{C}}^{-}\boldsymbol{\gamma}_i, \frac{1}{N}\sum_{i=1}^{N}\boldsymbol{\gamma}_i'(\bar{\mathbf{C}}^{-})'\bar{\boldsymbol{\Sigma}}_{\epsilon}\bar{\mathbf{C}}^{-}\mathbf{A}_i\right]', \\ \boldsymbol{b}_{2,b2sls} &= \lim_{N\to\infty}\left[\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{N}\sum_{l=1}^{N}g_{il}\boldsymbol{\beta}_0'(\mathbf{0}_{k\times 1}, \boldsymbol{\Sigma}_{v,l})\boldsymbol{\Delta}_{jl}^{-1\prime}\bar{\mathbf{C}}^{-}\boldsymbol{\gamma}_i, \frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{N}\boldsymbol{\gamma}_i'(\bar{\mathbf{C}}^{-})'\boldsymbol{\Delta}_{ji}^{-1}(\mathbf{0}_{k\times 1}, \boldsymbol{\Sigma}_{v,i})'\right]', \\ \boldsymbol{b}_{3,b2sls} &= \lim_{N\to\infty}\left[\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{N}\sum_{l=1}^{N}g_{il}\boldsymbol{\beta}_0'\mathbf{A}_l'(\bar{\mathbf{C}}^{-})'\boldsymbol{\Delta}_{ji}^{-1}\left(\sigma_i^2, \mathbf{0}_{1\times k}\right)', \frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{N}\left(\sigma_i^2, \mathbf{0}_{1\times k}\right)\boldsymbol{\Delta}_{ji}^{-1\prime}\bar{\mathbf{C}}^{-}\mathbf{A}_i\right]', \end{aligned} \tag{32}$$

g_{il} *is the* $(i,l)^{th}$ *element of* $\mathbf{G}_0$ *defined in (10),* $\bar{\mathbf{\Sigma}}_\epsilon$ *is given by (25),* $\bar{\mathbf{C}}^- = \bar{\mathbf{C}}'(\bar{\mathbf{C}}\bar{\mathbf{C}}')^{-1}$, *and* $\mathbf{\Delta}_{ji}^{-1}$ *is defined in (4).*

(c) *the B2SLS estimator is the best IV estimator if* $(N,T) \xrightarrow{j} \infty$ *with* $T/N \to 0$ *and if the errors* $\{e_{it}\}$ *are independently and identically distributed with mean zero and variance* σ_e^2.

First note that the identification of $\boldsymbol{\delta}_0$ is ensured by the nonsingularity of $(NT)^{-1}\mathbf{L}_0'\mathbf{M}_f^b\mathbf{L}_0$, which corresponds to the full column rank condition in Assumption 8(a) where the IV matrix $\mathbf{Q}$ is now replaced with the infeasible best IV $\mathbf{L}_0$. Second, as is the case with the 2SLS estimator, Theorem 1 shows that the B2SLS estimator is consistent if N is large, whether T is small or large; the B2SLS estimator is normally distributed if $\sqrt{T}/N \to 0$ as both N and T jointly tend to infinity. If $T/N \to 0$, $\sqrt{NT}(\hat{\boldsymbol{\delta}}_{b2sls} - \boldsymbol{\delta}_0) \sim N(\mathbf{0}, \mathbf{\Sigma}_{b2sls})$. However, if $T/N \to \kappa > 0$, there exists a bias term that involves $\sqrt{T/N}\boldsymbol{b}_{2sls}$. A similar procedure for bias-correcting the 2SLS estimator as described in the last section may be applied here, but we leave this topic for future research and focus on the inference when T is smaller than N in the present article.

Finally, it is worth noting that under Assumption 6, $(\mathbf{I}_N - \rho_0\mathbf{W})^{-1}\mathbf{X}_{.t}\boldsymbol{\beta}_0 = \sum_{r=1}^{N}\rho_0^r\mathbf{W}^r\mathbf{X}_{.t}\boldsymbol{\beta}_0$, which implies that the term $\mathbf{G}_0\mathbf{X}_{.t}\boldsymbol{\beta}_0$ can be approximated by linear combinations of $\mathbf{X}_{.t}\boldsymbol{\beta}$, $\mathbf{W}\mathbf{X}_{.t}\boldsymbol{\beta}$, $\mathbf{W}^2\mathbf{X}_{.t}\boldsymbol{\beta}$, ... Clearly, the higher the power of $\mathbf{W}$ included, the better the approximation. However, the efficiency gain by including more instruments may not be significant. In practice, the 2SLS estimator with instruments $(\mathbf{X}_{.t}, \mathbf{W}\mathbf{X}_{.t}, \mathbf{W}^2\mathbf{X}_{.t})$ is often found to perform well enough. The finite sample properties of $\hat{\boldsymbol{\delta}}_{b2sls}$ will be compared with those of $\hat{\boldsymbol{\delta}}_{2sls}$ using Monte Carlo techniques in Section 4.

3.3. GMM estimation

The third estimator we propose is the GMM estimator that utilizes quadratic moment conditions based on the properties of the idiosyncratic errors in addition to the 2SLS-type linear moments. The use of the quadratic moments for SAR models was proposed by Lee (2007) and later extended by Lin and Lee (2010) and Lee and Yu (2014). The first motivation for adopting the quadratic moments is to improve efficiency by exploiting the absence of cross-sectional correlation in the errors of the true model. The second motivation (and also benefit) is that it allows us to estimate the spatial coefficient when there are no exogenous regressors and hence no available instruments. In this section, we show that this idea can be extended to spatial models with common factors.

Specifically, we consider the following sample moment conditions, which consist of h quadratic moments and q linear moments:[22]

$$\mathbf{g}_{NT}(\boldsymbol{\delta}) = (NT)^{-1}\left[\boldsymbol{\xi}'(\boldsymbol{\delta})\mathbf{M}^b\mathbf{P}_1^b\mathbf{M}^b\boldsymbol{\xi}(\boldsymbol{\delta}), ..., \boldsymbol{\xi}'(\boldsymbol{\delta})\mathbf{M}^b\mathbf{P}_h^b\mathbf{M}^b\boldsymbol{\xi}(\boldsymbol{\delta}), \boldsymbol{\xi}'(\boldsymbol{\delta})\mathbf{M}^b\mathbf{Q}'\right]', \tag{33}$$

where $\boldsymbol{\xi}(\boldsymbol{\delta})$ is the vector of residuals given by

$$\boldsymbol{\xi}(\boldsymbol{\delta}) = [\mathbf{I}_T \otimes \mathbf{S}(\rho)]\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}, \tag{34}$$

and $\boldsymbol{\delta} = (\rho, \boldsymbol{\beta}')'$ represents the unknown parameters in the parameter space, Δ_{sp}. For $l = 1, 2, ..., h$, we define $\mathbf{P}_l^b = \mathbf{I}_T \otimes \mathbf{P}_l$, where $\mathbf{P}_l = (p_{l,ij})$ is an N-dimensional square matrix with zero diagonal, namely, $diag(\mathbf{P}_l) = (p_{l,11}, p_{l,22}, ..., p_{l,NN})' = 0$.

To grasp the idea behind the quadratic moments, it will be helpful to consider the simpler scenario where there are no common factors: the lth population quadratic moment at the true value $\boldsymbol{\delta}_0$ will be reduced to

[22]We use the aggregated moment conditions over time instead of a moment condition for each period separately, since the latter approach may induce the many-moment bias problem and is beyond the scope of the current article. See Lee and Yu (2014) for a discussion of this issue for spatial models.

$$E\left(\mathbf{e}'\mathbf{P}_l^b\mathbf{e}\right)=\sum_{i=1}^{N}\sum_{j=1}^{N}p_{l,ji}\sum_{t=1}^{T}E(e_{it}e_{jt})=\sum_{i=1}^{N}p_{l,ii}\sum_{t=1}^{T}E\left(e_{it}^2\right)=0,$$

where $p_{l,ji}$ is the $(j,i)^{th}$ element of matrix $\mathbf{P}_l$, the second equality follows from the uncorrelatedness between e_{it} and e_{jt} $(i\neq j)$, and the last equality follows from the assumption that $diag(\mathbf{P}_l)=0$. It is now clear that the $\mathbf{P}_l$ matrix is used to eliminate the variance of e_{it} in the weighted sums of all cross-section covariances in order to achieve zero expectation. We focus on the weighted sums because utilizing individual correlation coefficients would entail too many moment conditions. Also note that since we allow for unknown heteroskedasticity, we need all diagonal elements of $\mathbf{P}_l$ to be zero. By contrast, imposing $tr(\mathbf{P}_l)=0$ would be sufficient if e_{it} are homoskedastic.

The GMM estimator, $\hat{\boldsymbol{\delta}}_{gmm}$, is then defined as

$$\hat{\boldsymbol{\delta}}_{gmm}=\underset{\boldsymbol{\delta}\in\Delta_{sp}}{\operatorname{argmin}}\,\mathbf{g}_{NT}'(\boldsymbol{\delta})\mathbf{A}_{NT}^{w\prime}\mathbf{A}_{NT}^{w}\mathbf{g}_{NT}(\boldsymbol{\delta}),\tag{35}$$

where $\mathbf{g}_{NT}(\boldsymbol{\delta})$ is given by (33), and $\mathbf{A}_{NT}^{w}$ is a full row rank matrix of $k_a\times(h+q)$ dimension, where $k_a\geq k+1$, and $\mathbf{A}_{NT}^{w}$ is assumed to converge to a full rank nonrandom matrix $\mathbf{A}^{w}$. The following additional assumption is needed for the asymptotic analysis of the GMM estimator.

Assumption 9. *The $N\times N$ matrices $\boldsymbol{P}_l$, for $l=1,2,...,h$, used in the moment conditions given by (33), are nonstochastic with zero diagonal, and have bounded maximum row and column sum norms, namely, $||\mathbf{P}_l||_\infty<K$ and $||\mathbf{P}_l||_1<K$.*

The following theorem summarizes the asymptotic properties of the GMM estimator.

Theorem 3. *Consider the panel data model given by (1) and suppose that Assumptions 1–9 hold. Then,*

(a) *the GMM estimator, $\hat{\boldsymbol{\delta}}_{gmm}$, defined in (35), is consistent for $\boldsymbol{\delta}_0$ as $N\to\infty$, for T fixed or $T\to\infty$,*
(b) *as $(N,T)\xrightarrow{j}\infty$ with $\sqrt{T}/N\to 0$, we have*

$$\sqrt{NT}\left(\hat{\boldsymbol{\delta}}_{gmm}-\boldsymbol{\delta}_0\right)\sim\left(\mathbf{D}'\mathbf{A}^{w\prime}\mathbf{A}^{w}\mathbf{D}\right)^{-1}\mathbf{D}'\mathbf{A}^{w\prime}\mathbf{A}^{w}\sqrt{\frac{T}{N}}\boldsymbol{b}_{gmm}+N\left(\mathbf{0},\boldsymbol{\Sigma}_{gmm}\right),\tag{36}$$

where

$$\boldsymbol{\Sigma}_{gmm}=\left(\mathbf{D}'\mathbf{A}^{w\prime}\mathbf{A}^{w}\mathbf{D}\right)^{-1}\mathbf{D}'\mathbf{A}^{w\prime}\mathbf{A}^{w}\boldsymbol{\Sigma}_g\mathbf{A}^{w\prime}\mathbf{A}^{w}\mathbf{D}\left(\mathbf{D}'\mathbf{A}^{w\prime}\mathbf{A}^{w}\mathbf{D}\right)^{-1},\tag{37}$$

$$\begin{aligned}\mathbf{D}&=\left[\left(\mathbf{d}_p,\quad\mathbf{0}_{h\times k}\right)',\boldsymbol{\Psi}_{QML}'\right]',\boldsymbol{\Sigma}_g=\begin{pmatrix}\boldsymbol{\Sigma}_p & \mathbf{0}_{h\times q}\\ \mathbf{0}_{q\times h} & \boldsymbol{\Omega}_{QMe}\end{pmatrix},\\ \mathbf{d}_p&=\lim_{N\to\infty}N^{-1}\left(\sum_{i=1}^{N}\tilde{g}_{1,ii}^{s}\sigma_i^2,\sum_{i=1}^{N}\tilde{g}_{2,ii}^{s}\sigma_i^2,...,\sum_{i=1}^{N}\tilde{g}_{h,ii}^{s}\sigma_i^2\right)',\end{aligned}\tag{38}$$

where $\boldsymbol{\Psi}_{QML}$ and $\boldsymbol{\Omega}_{QMe}$ are given by (18) and (19), respectively, $\tilde{g}_{l,ii}^{s}$ $(l=1,2,...,h)$ is the ith diagonal element of $\tilde{\mathbf{G}}_l(\rho_0)=\mathbf{P}_l^{s}\mathbf{G}_0$, $\mathbf{P}_l^{s}=\mathbf{P}_l+\mathbf{P}_l'$, $\boldsymbol{\Sigma}_p$ is an $h\times h$ matrix of which the $(l,l')^{th}$ element is given by $\lim_{N\to\infty}N^{-1}tr\left[(\mathbf{P}_l\odot\mathbf{P}_{l'}^{s})\boldsymbol{\Sigma}_e\right]$, $\odot$ denotes the Hadamard (or entrywise) product, $\boldsymbol{\Sigma}_e$ is an $N\times N$ matrix of which the $(i,j)^{th}$ element is given by $\lim_{T\to\infty}T^{-1}tr(\boldsymbol{\Omega}_{e,i}\boldsymbol{\Omega}_{e,j})$,

$$\boldsymbol{b}_{gmm}=\left(\boldsymbol{b}_p',\boldsymbol{b}_{2sls}'\right)',\tag{39}$$

$\boldsymbol{b}_{2sls}$ *is given by (21),* $\boldsymbol{b}_p$ *is an* $h \times 1$ *vector of which the lth* $(l = 1, 2, ..., h)$ *element is given by*

$$b_l^p = \frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{N} p_{ji,l} tr\left[(\bar{\mathbf{C}}^-)'\bar{\boldsymbol{\Sigma}}_\epsilon \bar{\mathbf{C}}^- \boldsymbol{\gamma}_i\boldsymbol{\gamma}_j'\right] - \frac{2}{N}\sum_{i=1}^{N}\sum_{j=1}^{N}\sum_{r=1}^{N} p_{ri,l}\left(\sigma_i^2, \mathbf{0}_{1\times k}\right)\boldsymbol{\Delta}_{ji}^{-1\prime}\bar{\mathbf{C}}^-\boldsymbol{\gamma}_r,$$

$\bar{\boldsymbol{\Sigma}}_\epsilon$ *is given by (25),* $\bar{\mathbf{C}}^- = \bar{\mathbf{C}}'(\bar{\mathbf{C}}\bar{\mathbf{C}}')^{-1}$, *and* $\boldsymbol{\Delta}_{ji}^{-1}$ *is defined in (4).*

Theorem 3 shows that the consistency of the GMM estimator only requires N to be large, while its asymptotic normality was established under $(N, T)\xrightarrow{j}\infty$ with $\sqrt{T}/N \to 0$. The asymptotic bias in the case of $T/N \to \kappa > 0$ consists of the bias term, $\boldsymbol{b}_{2sls}$, due to the linear moments and the bias, $\boldsymbol{b}_p$, associated with the quadratic moments. As before, the asymptotic bias will vanish if $T/N \to 0$ and then $\sqrt{NT}(\hat{\boldsymbol{\delta}}_{gmm} - \boldsymbol{\delta}_0) \sim N(\mathbf{0}, \boldsymbol{\Sigma}_{gmm})$.

It may be helpful to examine the components of asymptotic variance matrix to gain some intuition behind the complex formula. $\boldsymbol{\Sigma}_p$ is the variance-covariance matrix of the quadratic moments and can be readily verified by applying Lemma S.9 in the Supporting Information. As $T^{-1}tr(\boldsymbol{\Omega}_{e,i}\boldsymbol{\Omega}_{e,j}) = T^{-1}\sum_{t=1}^{T}\sum_{s=1}^{T} E(e_{it}e_{is})E(e_{jt}e_{js})$, which would simplify to $\sigma_i^2\sigma_j^2$ if e_{it} are serially uncorrelated. The covariances between the quadratic and linear moments are zero due to $\mathbf{P}_l$ having zero diagonal.[23] The matrix $\mathbf{D}$ is the expectation of $-\partial\mathbf{g}_{NT}'(\boldsymbol{\delta})/\partial\boldsymbol{\delta}$ at $\boldsymbol{\delta}_0$. The component $\mathbf{d}_p$ originates from the nonzero correlation between the spatial variable $\mathbf{Y}^*$ and error $\mathbf{e}$, and such correlation only depends on ρ_0 in the limit. $\boldsymbol{\Psi}_{QML}$ and $\boldsymbol{\Omega}_{QMe}$ arise from the linear moments and are in accord with the discussion of the 2SLS estimation.

Remark 10. The identifiability of $\boldsymbol{\delta}_0$ under the GMM approach can be achieved via the linear moment conditions alone without the quadratic moments if the full column rank condition in Assumption 8(a) holds (see the proof of Theorem 3 in Appendix B). If one has to rely on the quadratic moments to identify $\boldsymbol{\delta}_0$, the same discussions on $\mathbf{P}_l$ for SAR models without common factors can be applied to identify ρ_0.[24] Once ρ_0 is identified, $\boldsymbol{\beta}_0$ can be identified if $(NT)^{-1}\mathbf{Q}'\mathbf{M}_f^b\mathbf{X}$ has the full rank k. Note that the presence of common factors does not help identify ρ_0, since they do not contain any cross-sectional variation. Also recall from Remark 9 that the common factors can be treated as observable when analyzing the identification condition since unknown factors can be consistently estimated with an error of order $O(1/\sqrt{N})$ under the assumptions.

We can obtain the (infeasible) efficient GMM estimator using the optimal weighting matrix, $\boldsymbol{\Sigma}_g^{-1}$, in the usual fashion, namely,

$$\hat{\boldsymbol{\delta}}_{gmm}^* = \underset{\boldsymbol{\delta}\in\Delta_{sp}}{\operatorname{argmin}}\, \mathbf{g}_{NT}'(\boldsymbol{\delta})\boldsymbol{\Sigma}_g^{-1}\mathbf{g}_{NT}(\boldsymbol{\delta}). \tag{40}$$

Then under the same assumptions of Theorem 3 as $(N, T)\xrightarrow{j}\infty$ with $\sqrt{T}/N \to 0$, we have

$$\sqrt{NT}\left(\hat{\boldsymbol{\delta}}_{gmm}^* - \boldsymbol{\delta}_0\right) \sim \left(\mathbf{D}'\boldsymbol{\Sigma}_g^{-1}\mathbf{D}\right)^{-1}\mathbf{D}'\boldsymbol{\Sigma}_g^{-1}\sqrt{\frac{T}{N}}\boldsymbol{b}_{gmm} + N\left(\mathbf{0}, \boldsymbol{\Sigma}_{gmm}^*\right),$$

which again implies that if $T/N \to 0$, there is no asymptotic bias and $\sqrt{NT}(\hat{\boldsymbol{\delta}}_{gmm}^* - \boldsymbol{\delta}_0) \sim N(\mathbf{0}, \boldsymbol{\Sigma}_{gmm}^*)$, where $\boldsymbol{\Sigma}_{gmm}^* = (\mathbf{D}'\boldsymbol{\Sigma}_g^{-1}\mathbf{D})^{-1}$.

A consistent estimator of $\boldsymbol{\Sigma}_{gmm}$, denoted by $\hat{\boldsymbol{\Sigma}}_{gmm}$,[25] can be obtained naturally by replacing $\mathbf{D}$ and $\boldsymbol{\Sigma}_g$ in (37) with $\hat{\mathbf{D}}$ and $\hat{\boldsymbol{\Sigma}}_g$, respectively, where

[23] See Lemma S.10 in the Supporting Information. The variance of the linear and quadratic form does not involve the third and fourth moments of the errors if $\mathbf{P}_l$ has zero diagonal.

[24] See, for example, Assumption 5(b) in Lin and Lee (2010). It can be readily modified to allow for serial dependence in the errors without much difficulty. Also see Proposition 1 of Pesaran and Yang (2020b).

[25] The proof that $\hat{\boldsymbol{\Sigma}}_{gmm}$ is consistent for $\boldsymbol{\Sigma}_{gmm}$ is given in the Supporting Information.

$$\hat{\mathbf{D}} = \left[\left(\hat{\mathbf{d}}_p, \quad \mathbf{0}_{r\times k}\right)', \hat{\mathbf{\Psi}}'_{QML}\right]', \hat{\mathbf{\Psi}}_{QML} = (NT)^{-1}\mathbf{Q}'\mathbf{M}^b\mathbf{L},$$

$$\hat{\mathbf{d}}_p = (NT)^{-1}\left(\sum_{i=1}^{N}\hat{\tilde{g}}^s_{1,ii}\hat{\mathbf{e}}'_{i.}\hat{\mathbf{e}}_{i.}, \sum_{i=1}^{N}\hat{\tilde{g}}^s_{2,ii}\hat{\mathbf{e}}'_{i.}\hat{\mathbf{e}}_{i.}, \ldots, \sum_{i=1}^{N}\hat{\tilde{g}}^s_{h,ii}\hat{\mathbf{e}}'_{i.}\hat{\mathbf{e}}_{i.}\right)',$$

$\hat{\mathbf{e}} = \mathbf{M}^b(\mathbf{Y} - \mathbf{L}\hat{\boldsymbol{\delta}}_{gmm})$, $\hat{\tilde{g}}^s_{l,ii}$ is the ith diagonal element of $\tilde{\mathbf{G}}_l(\hat{\rho})$, and

$$\hat{\mathbf{\Sigma}}_g = \begin{pmatrix} \hat{\mathbf{\Sigma}}_p & \mathbf{0}_{h\times q} \\ \mathbf{0}_{q\times h} & \hat{\mathbf{\Omega}}_{QMe} \end{pmatrix},$$

where $\hat{\mathbf{\Omega}}_{QMe}$ is the Newey-West type estimator defined analogously to (27), $\hat{\mathbf{\Sigma}}_p$ is an $h \times h$ matrix of which the $(l,l')^{th}$ element is given by $(NT)^{-1}\sum_{i=1}^{N}\sum_{j=1}^{N} p_{l,ji}(p_{l',ij} + p_{l',ji})\hat{\varsigma}_{e,ij}$, with

$$\hat{\varsigma}_{e,ij} = T\hat{\gamma}_{e,i}(0)\hat{\gamma}_{e,j}(0) + 2\sum_{\kappa=1}^{M_\kappa}(T-\kappa)\left(1 - \frac{\kappa}{M_\kappa+1}\right)\hat{\gamma}_{e,i}(\kappa)\hat{\gamma}_{e,j}(\kappa), \tag{41}$$

$\hat{\gamma}_{e,i}(\kappa) = T^{-1}\sum_{t=\kappa+1}^{T}\hat{e}_{it}\hat{e}_{i,t-\kappa}$, $\hat{\gamma}_{e,j}(\kappa)$ is defined likewise, and M_κ is the maximum lag length. Similarly, we can estimate $\mathbf{\Sigma}^*_{gmm}$ by $\hat{\mathbf{\Sigma}}^*_{gmm} = (\hat{\mathbf{D}}^{*\prime}\hat{\mathbf{\Sigma}}^{*-1}_g\hat{\mathbf{D}}^*)^{-1}$, where $\hat{\mathbf{D}}^*$ and $\hat{\mathbf{\Sigma}}^*_g$ will be computed using $\hat{\mathbf{e}}^* = \mathbf{M}^b(\mathbf{Y} - \mathbf{L}\hat{\boldsymbol{\delta}}^*_{gmm})$ instead of $\hat{\mathbf{e}}$.

Lastly, we conclude this section by discussing the choice of $\mathbf{P}_l$ for the quadratic moments. Theorem 3 shows that the consistency of the GMM estimator is not affected by the choice of $\mathbf{P}_l$ as long as they satisfy Assumption 9, but the efficiency property obviously depends on $\mathbf{P}_l$. One may hope to select the best $\mathbf{P}_l$ that leads to minimum variance, but unfortunately such optimal choice in general is not available in the presence of unknown heteroskedastic and serially correlated errors. This can be seen from the precision matrix of the efficient GMM estimator:

$$\mathbf{D}'\mathbf{\Sigma}_g^{-1}\mathbf{D} = \begin{pmatrix} \mathbf{d}'_p\mathbf{\Sigma}_p^{-1}\mathbf{d}_p & \mathbf{0}_{1\times k} \\ \mathbf{0}_{k\times 1} & \mathbf{0}_{k\times k} \end{pmatrix} + \frac{1}{NT}\left(\mathbf{Q}'\mathbf{M}^b_f\mathbf{L}_0\right)'\mathbf{\Omega}_{QMe}\left(\mathbf{Q}'\mathbf{M}^b_f\mathbf{L}_0\right).$$

Ideally, one wish to find $\mathbf{P}_l$ to maximize $\mathbf{d}'_p\mathbf{\Sigma}_p^{-1}\mathbf{d}_p$, but this term depends on the unknown covariance structure of the errors. In the special case where the errors are independent and identically distributed (i.i.d.), it is known in the spatial literature that the best $\mathbf{P}_l$ within the class of matrices under Assumption 9 is given by $\mathbf{P}^* = \mathbf{G}_0 - Diag(\mathbf{G}_0)$ (Lee, 2007; Lee and Yu, 2014).[26] This result can be extended to our model with common factors by similar reasoning as that for the B2SLS estimator. Nonetheless, the i.i.d. assumption is too restrictive in reality and hence $\mathbf{P}^*$ may not be optimal. On a positive note, a convenient good candidate for $\mathbf{P}_l$ is simply $\mathbf{W}$, which clearly satisfies Assumption 9. Since $\mathbf{W}^2$ and higher powers of $\mathbf{W}$ also have bounded row and column sum norms, they will be valid candidates too if their diagonal elements are set to zero.[27] However, including higher powers of $\mathbf{W}$ often do not lead to notable efficiency gain in practice. Therefore, $\mathbf{P}_1 = \mathbf{W}$ and $\mathbf{P}_2 = \mathbf{W}^2 - Diag(\mathbf{W}^2)$ are commonly adopted and found to have adequate performance in many applications. Moreover, they do not depend on any unknown parameter as $\mathbf{P}^*$ does and thus do not require a preliminary estimation procedure. Henceforth, we will present GMM estimation results using $\mathbf{P}_1$ and $\mathbf{P}_2$ in Monte Carlo simulations and the empirical application, but these results remain robust to including more quadratic moments based on higher powers of $\mathbf{W}$.

[26]Since $\mathbf{G}_0$ involves unknown parameter ρ_0, it may be constructed from some initial consistent estimate of ρ_0 obtained in a preliminary estimation step.

[27]As in the case of B2SLS estimation, under Assumption 6, $\mathbf{G}_0 = \mathbf{W}(\mathbf{I}_N - \rho_0\mathbf{W})^{-1} = \mathbf{W} + \rho_0\mathbf{W}^2 + \rho_0^2\mathbf{W}^3 + \ldots$, and therefore including higher powers of $\mathbf{W}$ as $\mathbf{P}_l$ can be seen as an approximation for $\mathbf{P}^*$.

4. Monte Carlo experiments

This section provides Monte Carlo evidence on the finite sample properties of the proposed estimators under various specifications of the disturbance process and under different intensities of spatial dependence. It also compares the performance of the proposed estimators with that of alternative estimators. In specific, we consider the following Data Generating Process (DGP), which adds spatial dependence to the Monte Carlo design in Pesaran (2006):

$$\begin{aligned} y_{it} &= \rho y_{it}^{*} + \beta_1 x_{it1} + \beta_2 x_{it2} + \boldsymbol{\gamma}_{y,i}' \boldsymbol{f}_t + e_{it}, \\ x_{itp} &= \boldsymbol{\gamma}_{x,ip}' \boldsymbol{f}_t + \upsilon_{itp}, p = 1, 2, \end{aligned} \tag{42}$$

for $i = 1, 2, ..., N,$ and $t = 1, 2, ..., T$. The unobserved factors are generated by

$$\begin{aligned} f_{lt} &= \rho_{fl} f_{l,t-1} + \varsigma_{f_{lt}}, l = 1, 2, ..., m; t = -49, -48, ..., 0, 1, ..., T, \\ \varsigma_{f_{lt}} &\sim IIDN\left(0, 1 - \rho_{fl}^2\right), \rho_{fl} = 0.5, f_{l,-50} = 0, \end{aligned}$$

where the first 50 observations are discarded. The factor loadings are assumed to be $\gamma_{y,i1} \sim IIDN(1, 0.2), \gamma_{y,i2} \sim IIDN(1, 0.2),$ and

$$\begin{pmatrix} \gamma_{x,i11} & \gamma_{x,i12} \\ \gamma_{x,i21} & \gamma_{x,i22} \end{pmatrix} \sim IID \begin{pmatrix} N(0.5, 0.5) & N(0, 0.5) \\ N(0, 0.5) & N(0.5, 0.5) \end{pmatrix}.$$

The idiosyncratic errors of the x_{itp} processes, $(\upsilon_{it1}, \upsilon_{it2})',$ are generated as

$$\begin{aligned} \upsilon_{it,p} &= \rho_{\upsilon_{ip}} \upsilon_{it-1,p} + \vartheta_{it,p}, t = -49, -48, ..., 0, 1, ..., T, \\ \vartheta_{it,p} &\sim N\left(0, 1 - \rho_{\vartheta_{ip}}^2\right), \upsilon_{ip,-50} = 0, \\ \rho_{\vartheta_{ip}} &\sim IIDU(0.05, 0.95), p = 1, 2, \end{aligned}$$

where the first 50 observations are discarded.

We consider two different designs for the idiosyncratic errors of y_{it}:[28]

- The errors e_{it} are generated from $IIDN(0, 1)$. The main goal of this baseline setup is to compare the efficiency properties of the competing estimators. In particular, it is of interest to examine if the B2SLS and GMM estimators are more efficient than the 2SLS estimator.
- The errors e_{it} are serially correlated and heteroskedastic. In particular, they are specified as AR(1) processes for the first half of individual units and as MA(1) processes for the remaining half:

$$e_{it} = \rho_{ie} e_{i,t-1} + \sigma_i \left(1 - \rho_{ie}^2\right)^{1/2} \zeta_{it}, i = 1, 2, ..., \lfloor N/2 \rfloor, \tag{43}$$

$$\begin{aligned} e_{it} &= \sigma_i \left(1 + \theta_{ie}^2\right)^{1/2} (\zeta_{it} + \theta_{ie} \zeta_{i,t-1}), i = \lfloor N/2 \rfloor + 1, \lfloor N/2 \rfloor + 2, ..., N, \\ \zeta_{it} &\sim IIDN(0, 1), \sigma_i^2 \sim IIDU(0.5, 1.5), \\ \rho_{ie} &\sim IIDU(0.05, 0.95), e_{i,-50} = 0, \end{aligned} \tag{44}$$

where $\lfloor a \rfloor$ denotes the integer part of a number a.

The spatial weights matrix is specified as the q-ahead-and-q-behind circular neighbors weights matrix, which are commonly employed in the literature. A q-ahead-and-q-behind matrix is motivated to capture spatial relations in which all units are located in a circle; the q units immediately

[28]We have also examined the case where the errors are independent over time and heteroskedastic across individual units. The results are presented in the Supporting Information.

ahead of and behind a particular unit are considered "neighbors" and assigned equal weights. For example, for the 2-ahead-and-2-behind spatial matrix, the ith row of $\mathbf{W}$ has nonzero elements in the positions $i-2, i-1, i+1, i+2$, and each weighs 1/4 due to row normalization. Without loss of generality, we report the results for $q=1$; other values of q lead to similar results and are not presented to save space. In all experiments, the true number of factors is $m=2$, and the true parameter values are $\rho=0.4, \beta_1=1$, and $\beta_2=2$.[29] The sample sizes are combinations of $N=30$, 50, 100, 500, 1,000 and $T=20$, 30, 50, 100. Each experiment is replicated 2,000 times.

The parameters of interest for model (42) are $(\rho, \beta_1, \beta_2)'$, which are estimated by the following methods:

- The naive 2SLS estimator, which ignores the latent factors and applies a standard 2SLS estimation procedure directly with instruments $\mathbf{Q}_{.t}^{[2]}=(\mathbf{X}_{.t}, \mathbf{W}\mathbf{X}_{.t}, \mathbf{W}^2\mathbf{X}_{.t})$, for $t=1,2,...,T$, where the superscript of $\mathbf{Q}_{.t}$ denotes that the highest power of $\mathbf{W}$ used in constructing the instruments.
- The infeasible 2SLS estimator, which assumes the factors are known and utilizes instruments $\mathbf{Q}_{.t}^{[2]}$, for $t=1,2,...,T$.
- The 2SLS estimator given by (14) with instruments $\mathbf{Q}_{.t}^{[2]}$, for $t=1,2,...,T$.
- The B2SLS estimator given by (28), which is implemented in two steps. In the first step, we compute a preliminary 2SLS estimate following (14) using instruments $\mathbf{Q}_{.t}^{[2]}$, for $t=1,2,...,T$. In the second step, the B2SLS estimate is obtained by using the estimated best IV matrix, $\left[(\mathbf{I}_T \otimes \hat{\mathbf{G}})\mathbf{X}\hat{\boldsymbol{\beta}}_{2sls}, \mathbf{X}\right]$, where $\hat{\mathbf{G}}=\mathbf{G}(\hat{\rho}_{2sls})$, in (28).
- The efficient GMM estimator given by (40) that uses $\mathbf{P}_1=\mathbf{W}$ and $\mathbf{P}_2=\mathbf{W}^2-Diag(\mathbf{W}^2)$ in the quadratic moments and $\mathbf{Q}_{.t}^{[2]}$ as IVs in the linear moments. It is obtained by a two-step procedure. In the first step, we take the identity matrix as the moments weighting matrix and compute a preliminary GMM estimate. In the second step, the estimated inverse of the covariance of moments is used as the weighting matrix, and the model is re-estimated using the same $\mathbf{P}_1, \mathbf{P}_2$, and IVs.
- The MLE developed by Bai and Li (2014). This procedure assumes that the disturbances of the model are independently distributed with heteroskedastic variances and explicitly estimates all of the heteroskedasticity and factor loadings. It is important to note that the asymptotic distribution of the MLE was derived under the assumption that $N, T \to \infty$ and $\sqrt{N}/T \to 0$. The incidental parameters in the time dimension are avoided by estimating the sample variance of the factors rather than individual factors.[30] We compute the MLE following the Expectation-Maximization (EM) algorithm suggested by Bai and Li (2014). The number of factors is assumed known in the experiments to reduce the computational burden.[31] The size and power properties of the MLE are not reported in their paper.

For the robust variance estimation of the above methods (except the MLE), the Bartlett window width is chosen to be $\lfloor 2\sqrt{T} \rfloor$.[32]

Tables 1–4 collect the results of the experiments. Each table reports the estimates of bias, root mean squared error (RMSE), size, and power for the aforementioned estimators. Tables 1 and 3 report the results for the spatial coefficient, ρ; and 2 and 4 report the results for the slope

[29]We have also considered $\rho=0.8$, which represents high intensity of spatial dependence. The results are provided in the Supporting Information.

[30]Bai and Li (2014) point out that one could switch the role of N and T if T is much smaller than N. We do not report results under this interchange, since it requires different strict assumptions on the disturbances and does not improve the performance of MLE under our Monte Carlo designs.

[31]Bai and Li (2014) propose using an information criterion to estimate the number of factors in their Monte Carlo experiments.

[32]We also considered $\lfloor T^{1/3} \rfloor$ as the window size. The results are close but using $\lfloor 2\sqrt{T} \rfloor$ has slightly better size properties.

coefficient, β. The results for β_2 are similar to those for β_1 and thus omitted to save space. The results of the naive estimator are only presented in Tables 1 and 2, since ignoring the factors produces enormous biases and variances in all experiments, as expected.

We first observe that the 2SLS estimator exhibits very small biases and declining RMSEs as N and/or T increase. A comparison between the 2SLS and the infeasible 2SLS estimators suggests that the efficiency loss from using cross-sectional averages to approach the unobserved factors is quite small, almost indiscernible when the sample size is large. The B2SLS estimator is only marginally more efficient than the 2SLS estimator for the spatial parameter ρ when N is small, and it provides little or no improvement for the slope parameter $\boldsymbol{\beta}$. This implies that the IV matrix $\mathbf{Q}_{.t}^{[2]} = (\mathbf{X}_{.t}, \mathbf{W}\mathbf{X}_{.t}, \mathbf{W}^2\mathbf{X}_{.t})$ used in computing the 2SLS estimates approximates the best IV quite well in our experimental designs. The GMM estimator of ρ outperforms the 2SLS and B2SLS estimator in reducing the RMSEs, and it even beats the infeasible 2SLS estimator for modest to large sample size ($N \geq 100$). Finally, the MLE developed by Bai and Li (2014) produces the smallest RMSEs among all estimation methods, and the improvement for ρ is especially notable. Nonetheless, its computation for large values of N and T is rather strenuous, and its performance could be weakened if the number of factors is estimated, especially when the estimated number of factors is smaller than the true value.

Turning to size and power properties, as anticipated by the theoretical findings, the proposed estimators have good power and empirical sizes that are close to the 5% nominal size for large N and small to modest T, irrespective of whether the errors are heteroskedastic and serially correlated. In cases where N is much smaller than T, the rejection frequencies under the null of the 2SLS and B2SLS estimators are slightly higher than 5%, and the GMM estimator is more oversized than the 2SLS estimators. It is also evident that the size distortion is more pronounced for the spatial parameter than for the slope coefficients. In view of these results, it is worthwhile to bear in mind that the variance estimators cannot be applied to the small N large T scenarios. Also note that for the case where N and T are comparable, the asymptotic biases as shown in the theory were uncorrected. In contrast, the MLE performs well when the errors are independent; it has higher power than the other estimators and proper sizes close to the 5% nominal level when N is not too large relative to T. However, as its theory does not permit the presence of serial correlation in the errors, the MLE based tests are significantly over-sized in this case. For the combinations of N and T considered, the empirical sizes of the MLE range from 13% to 29%.

In summary, the proposed estimators exhibit robust performance to unknown heteroskedasticity and serial correlation in the errors. Furthermore, the estimators are also robust to different intensity of spatial dependence, as supported by the additional simulation results in the Supporting Information.

5. An empirical application to US house prices

In this section, we apply the proposed estimation methods to analyzing the spatial dependence of real house price changes in the US at the level of MSAs. Since neighboring regions are often influenced by the same aggregate supply and demand shocks, it is the purpose of this exercise to properly assess the strength of the spatial interconnections while netting out the effects of common factors. As we will see, the degree of spatial dependence will be exaggerated if the unobserved common effects are not effectively removed. In addition, we are also interested in the effects of possible determinant variables on house price growth, including both direct and indirect (spillover) effects.[33]

[33]Cohen, Ioannides, and Thanapisitikul (2016) and Bailey, Holly, and Pesaran (2016) focus on house price series itself and do not consider any explanatory variables.

Table 1. Small sample properties of estimators for the spatial parameter ρ ($\rho = 0.4$, i.i.d. errors).

	Bias (×100)				RMSE (×100)				Size (×100)				Power (×100)			
$N\backslash T$	20	30	50	100	20	30	50	100	20	30	50	100	20	30	50	100
	Naive 2SLS estimator (excluding factors)															
30	16.06	16.21	16.34	16.41	16.58	16.61	16.65	16.66	99.40	99.65	99.95	99.95	99.65	99.85	99.95	100.00
50	16.04	16.23	16.34	16.38	16.44	16.52	16.55	16.55	99.90	100.00	100.00	100.00	100.00	100.00	100.00	100.00
100	16.01	16.24	16.32	16.38	16.33	16.46	16.47	16.48	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
500	15.95	16.14	16.27	16.33	16.21	16.30	16.37	16.39	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
1,000	15.95	16.14	16.27	16.34	16.20	16.30	16.37	16.40	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
	Infeasible 2SLS estimator (including factors)															
30	−0.10	−0.04	0.00	0.01	2.40	1.91	1.47	0.99	5.10	4.95	5.15	4.45	13.50	18.15	29.10	52.10
50	0.04	0.05	0.00	0.00	1.85	1.48	1.11	0.77	5.75	5.40	5.25	5.15	20.45	29.70	43.00	72.25
100	0.01	0.01	0.01	0.01	1.30	1.03	0.78	0.55	5.25	4.35	4.35	4.40	34.75	48.50	71.15	95.20
500	−0.02	−0.01	0.00	0.00	0.59	0.46	0.35	0.25	5.60	4.85	4.65	4.50	92.60	98.75	100.00	100.00
1,000	0.00	0.00	−0.01	0.00	0.42	0.33	0.25	0.18	4.90	5.65	5.30	5.70	99.80	99.95	100.00	100.00
	2SLS estimator															
30	−0.08	−0.01	0.00	0.01	2.75	2.16	1.64	1.15	6.10	6.45	7.30	8.10	12.40	18.35	28.30	48.30
50	0.02	0.05	0.01	0.00	1.99	1.58	1.20	0.83	5.35	5.95	5.30	5.70	18.05	26.15	41.55	71.00
100	0.01	0.00	0.01	0.01	1.38	1.08	0.81	0.56	4.50	5.10	4.45	5.20	30.25	45.30	69.40	94.10
500	−0.02	−0.01	0.00	0.00	0.62	0.48	0.36	0.25	5.00	4.55	4.95	4.75	89.05	97.95	100.00	100.00
1,000	0.00	0.00	−0.01	0.00	0.44	0.34	0.26	0.18	4.70	4.90	5.20	5.40	99.45	99.95	100.00	100.00
	B2SLS estimator															
30	−0.12	−0.03	−0.01	0.00	2.74	2.16	1.63	1.15	6.00	6.40	7.10	8.00	12.25	18.40	27.90	48.05
50	0.00	0.03	0.00	0.00	1.99	1.58	1.19	0.83	5.30	6.10	5.25	5.45	18.10	25.75	41.30	70.90
100	−0.01	−0.01	0.01	0.01	1.38	1.08	0.80	0.56	4.75	5.05	4.45	5.20	29.75	45.10	69.20	94.15
500	−0.02	−0.01	0.00	0.00	0.62	0.48	0.36	0.25	4.95	4.55	4.85	4.60	88.65	98.05	100.00	100.00
1,000	0.00	0.00	−0.01	0.00	0.44	0.34	0.25	0.18	4.50	4.65	5.05	5.60	99.45	99.95	100.00	100.00
	GMM estimator															
30	−1.25	−1.11	−1.07	−1.02	2.60	2.11	1.76	1.41	10.30	11.45	16.00	24.20	8.75	10.85	14.10	23.15
50	−0.69	−0.64	−0.64	−0.60	1.86	1.52	1.22	0.94	8.50	9.80	12.15	16.00	15.80	21.95	32.10	54.25
100	−0.33	−0.32	−0.31	−0.29	1.24	0.98	0.75	0.57	6.90	6.85	7.05	9.95	33.60	47.30	69.85	94.85
500	−0.08	−0.07	−0.07	−0.06	0.52	0.41	0.31	0.22	6.00	5.20	6.00	6.65	96.15	99.80	100.00	100.00
1,000	−0.03	−0.03	−0.04	−0.03	0.36	0.29	0.22	0.15	5.25	5.60	5.65	6.25	100.00	100.00	100.00	100.00
	MLE															
30	0.30	0.23	0.18	0.16	2.32	1.79	1.36	0.92	11.80	10.20	8.65	7.85	30.65	34.95	46.80	71.20
50	0.35	0.23	0.14	0.11	1.79	1.39	1.02	0.70	12.45	10.00	8.30	7.30	41.45	49.05	64.00	88.70
100	0.29	0.17	0.11	0.09	1.26	0.95	0.70	0.49	11.55	9.25	7.00	6.95	59.75	71.85	89.05	99.25
500	0.22	0.11	0.05	0.04	0.59	0.43	0.31	0.22	13.40	9.30	7.20	7.70	99.00	100.00	100.00	100.00
1,000	0.20	0.11	0.06	0.04	0.42	0.32	0.23	0.16	14.40	11.10	8.40	7.00	100.00	100.00	100.00	100.00

Notes: The DGP is given by (42), where $e_{it} \sim IIDN(0,1)$. The true parameter values are $\rho = 0.4$, $\beta_1 = 1$, and $\beta_2 = 2$. The naive estimator ignores latent factors, and the infeasible estimator treats factors as known. The naive 2SLS, infeasible 2SLS, and 2SLS estimators are computed using instruments $\mathbf{Q}_{.t}^{[2]} = (\mathbf{X}_{.t}, \mathbf{W}\mathbf{X}_{.t}, \mathbf{W}^2\mathbf{X}_{.t})$, for $t = 1, 2, ..., T$. The best 2SLS (B2SLS) estimator is computed using $[(\mathbf{I}_T \otimes \hat{\mathbf{G}})\mathbf{X}\hat{\beta}_{2sls}, \mathbf{X}]$, where $\hat{\mathbf{G}} = \mathbf{G}(\hat{\rho}_{2sls})$. The efficient two-step GMM estimator utilizes $\mathbf{P}_1 = \mathbf{W}$ and $\mathbf{P}_2 = \mathbf{W}^2 - Diag(\mathbf{W}^2)$ in the quadratic moments and $\mathbf{Q}_t^{[2]}$ in the linear moments. The MLE is computed by the EM algorithm described in Bai and Li (2014), assuming the number of factors is known. The number of replications is 2,000. The power is computed under $H_1 : \rho = 0.38$.

Table 2. Small sample properties of estimators for the slope parameter β_1 ($\beta_1 = 1$, i.i.d. errors).

	Bias (×100)				RMSE (×100)				Size (×100)				Power (×100)			
$M\backslash T$	20	30	50	100	20	30	50	100	20	30	50	100	20	30	50	100
	Naive 2SLS estimator (excluding factors)															
30	8.82	9.09	9.11	9.24	11.71	11.56	11.23	11.10	53.55	63.40	72.35	83.00	76.15	83.80	90.45	95.65
50	8.77	8.88	9.05	9.25	10.91	10.60	10.45	10.40	65.15	74.30	84.40	91.25	87.45	92.60	96.50	99.05
100	9.03	9.13	9.22	9.42	10.43	10.28	10.11	10.14	79.90	86.45	93.35	97.80	96.70	98.60	99.70	99.90
500	9.15	9.27	9.34	9.53	9.93	9.85	9.73	9.78	97.00	98.90	99.80	100.00	99.85	100.00	100.00	100.00
1,000	9.17	9.30	9.36	9.55	9.87	9.80	9.69	9.74	98.15	99.70	99.95	100.00	100.00	100.00	100.00	100.00
	Infeasible 2SLS estimator (including factors)															
30	0.05	0.02	−0.01	−0.01	4.50	3.57	2.73	1.88	5.50	5.20	5.40	5.10	21.80	30.45	47.85	75.95
50	−0.19	−0.17	−0.11	−0.09	3.45	2.66	2.01	1.40	5.45	4.40	4.65	4.50	28.55	42.75	66.25	92.75
100	−0.13	−0.04	−0.05	−0.05	2.46	1.92	1.48	1.01	5.65	5.65	5.35	5.05	52.45	73.75	91.90	99.85
500	−0.06	−0.04	−0.02	−0.01	1.07	0.84	0.66	0.47	4.85	4.35	5.20	5.80	99.80	100.00	100.00	100.00
1,000	0.01	0.01	0.01	0.00	0.79	0.62	0.47	0.33	6.10	6.00	5.30	5.30	100.00	100.00	100.00	100.00
	2SLS estimator															
30	0.06	0.04	0.03	0.03	4.73	3.77	2.91	2.05	5.75	6.30	7.35	7.35	20.00	29.20	47.00	75.60
50	−0.19	−0.18	−0.09	−0.09	3.61	2.76	2.08	1.45	4.70	5.10	4.75	4.85	26.05	39.60	65.80	92.60
100	−0.13	−0.05	−0.06	−0.05	2.54	1.96	1.51	1.03	5.40	5.05	4.75	5.05	46.85	70.70	90.80	99.85
500	−0.07	−0.05	−0.02	−0.01	1.12	0.86	0.67	0.48	4.15	4.30	5.00	5.30	99.25	100.00	100.00	100.00
1,000	0.02	0.01	0.01	0.00	0.82	0.64	0.48	0.33	5.45	5.20	5.25	5.25	100.00	100.00	100.00	100.00
	B2SLS estimator															
30	0.07	0.04	0.04	0.03	4.73	3.77	2.91	2.05	5.75	6.25	7.40	7.35	19.95	29.25	47.20	75.45
50	−0.19	−0.18	−0.09	−0.09	3.61	2.76	2.08	1.45	4.70	5.10	4.70	4.85	26.10	39.75	65.75	92.70
100	−0.13	−0.05	−0.06	−0.04	2.54	1.96	1.51	1.03	5.35	5.10	4.75	5.10	46.90	70.75	90.75	99.85
500	−0.07	−0.05	−0.02	−0.01	1.12	0.86	0.67	0.48	4.15	4.25	5.00	5.30	99.25	100.00	100.00	100.00
1,000	0.02	0.01	0.01	0.00	0.82	0.64	0.48	0.33	5.45	5.30	5.25	5.25	100.00	100.00	100.00	100.00
	GMM estimator															
30	0.16	0.17	0.17	0.17	4.79	3.80	2.92	2.06	5.70	6.95	7.30	7.55	21.35	30.70	49.25	77.55
50	−0.09	−0.08	0.01	0.01	3.64	2.77	2.07	1.45	4.75	4.90	4.80	4.85	27.90	41.60	67.25	93.55
100	−0.08	0.01	0.00	0.01	2.54	1.96	1.50	1.03	5.45	5.05	4.80	4.45	47.70	71.95	90.95	99.85
500	−0.06	−0.04	−0.01	0.00	1.11	0.86	0.67	0.47	4.10	4.30	5.00	5.30	99.20	100.00	100.00	100.00
1,000	0.02	0.02	0.02	0.01	0.81	0.64	0.48	0.33	5.30	5.40	5.50	5.40	100.00	100.00	100.00	100.00
	MLE															
30	−0.01	−0.01	−0.06	−0.04	5.05	3.85	2.86	1.93	11.45	9.30	7.70	6.10	29.80	35.10	48.45	75.80
50	−0.20	−0.16	−0.13	−0.11	3.76	2.79	2.06	1.43	10.20	7.00	5.95	5.55	36.20	47.15	68.15	93.15
100	−0.14	−0.05	−0.07	−0.06	2.68	2.01	1.52	1.04	10.60	8.00	6.80	6.05	58.45	76.55	91.85	99.90
500	−0.02	−0.01	−0.01	−0.01	1.18	0.88	0.67	0.48	10.70	6.50	5.90	7.00	99.60	100.00	100.00	100.00
1,000	0.04	0.00	0.02	−0.01	0.84	0.65	0.47	0.33	10.40	8.40	6.30	5.50	100.00	100.00	100.00	100.00

Notes: The DGP is given by (42), where $e_{it} \sim IIDN(0, 1)$. The true parameter values are $\rho = 0.4$, $\beta_1 = 1$, and $\beta_2 = 2$. The power is computed under $H_1 : \beta_1 = 0.95$. See also the notes to Table 1.

Table 3. Small sample properties of estimators for the spatial parameter ρ ($\rho = 0.4$, serially correlated and heteroskedastic errors).

	Bias (×100)				RMSE (×100)				Size (×100)				Power (×100)			
$N\backslash T$	20	30	50	100	20	30	50	100	20	30	50	100	20	30	50	100
	Infeasible 2SLS estimator (including factors)															
30	−0.16	−0.08	−0.01	0.00	2.85	2.34	1.81	1.26	5.85	7.00	6.65	5.85	13.60	16.95	23.20	37.55
50	0.07	0.05	−0.01	0.01	2.18	1.77	1.39	0.97	6.75	6.00	5.70	5.90	19.30	25.05	34.70	56.65
100	0.05	0.03	0.03	0.02	1.58	1.28	0.98	0.70	6.70	5.90	5.45	5.85	30.50	40.05	57.35	83.00
500	−0.04	−0.03	−0.02	−0.02	0.70	0.56	0.44	0.31	6.15	5.90	5.45	5.05	81.30	94.70	99.65	100.00
1,000	−0.01	−0.01	−0.01	0.00	0.50	0.40	0.31	0.22	6.05	5.95	6.40	6.75	98.45	99.95	100.00	100.00
	2SLS estimator															
30	−0.14	−0.07	−0.02	0.00	3.10	2.51	1.95	1.39	6.55	7.05	7.35	7.90	12.65	17.80	23.30	36.15
50	0.07	0.06	0.01	0.02	2.27	1.85	1.44	1.01	6.00	6.95	6.20	6.00	17.00	23.60	35.05	55.90
100	0.04	0.03	0.03	0.02	1.62	1.30	0.99	0.71	6.10	5.90	5.55	5.95	26.65	37.30	56.65	83.00
500	−0.04	−0.03	−0.02	−0.02	0.71	0.57	0.44	0.31	5.45	5.70	5.90	5.40	78.90	93.10	99.65	100.00
1,000	−0.01	−0.01	−0.01	0.00	0.51	0.40	0.31	0.23	5.65	5.25	6.05	6.55	97.65	99.95	100.00	100.00
	B2SLS estimator															
30	−0.19	−0.10	−0.04	−0.01	3.10	2.52	1.94	1.39	6.85	7.30	7.45	8.05	12.30	17.10	22.90	35.85
50	0.03	0.04	0.00	0.01	2.27	1.84	1.44	1.01	5.90	6.70	6.20	5.90	16.90	23.10	34.95	55.50
100	0.02	0.02	0.03	0.02	1.61	1.30	0.99	0.71	6.35	5.90	5.45	5.65	26.85	37.00	56.35	82.85
500	−0.05	−0.03	−0.02	−0.02	0.71	0.57	0.45	0.31	5.50	6.05	6.25	5.40	78.60	93.15	99.70	100.00
1,000	−0.01	−0.01	−0.01	0.00	0.51	0.40	0.31	0.23	5.60	5.15	5.95	6.30	97.70	99.95	100.00	100.00
	GMM estimator															
30	−1.11	−1.01	−0.97	−0.97	2.85	2.38	1.96	1.54	9.70	11.00	13.45	17.40	9.20	11.95	14.55	19.10
50	−0.54	−0.52	−0.55	−0.55	2.04	1.69	1.37	1.03	8.00	8.30	8.90	11.00	14.25	20.25	27.25	42.50
100	−0.24	−0.25	−0.25	−0.26	1.41	1.14	0.88	0.66	6.85	6.70	6.95	8.35	27.40	38.30	57.70	82.70
500	−0.08	−0.08	−0.07	−0.07	0.61	0.49	0.38	0.27	5.85	5.15	5.95	5.60	89.30	98.25	100.00	100.00
1,000	−0.04	−0.03	−0.04	−0.03	0.43	0.34	0.27	0.19	4.90	5.05	5.80	6.25	99.50	99.95	100.00	100.00
	MLE															
30	0.38	0.22	0.19	0.15	2.63	2.08	1.59	1.09	21.20	19.20	17.50	16.15	39.90	42.15	50.30	71.25
50	0.45	0.25	0.14	0.13	2.02	1.57	1.19	0.84	22.10	18.05	15.50	15.70	50.95	55.55	65.05	86.25
100	0.39	0.22	0.14	0.10	1.46	1.13	0.85	0.59	22.55	18.80	15.80	15.35	66.60	74.95	87.60	98.50
500	0.27	0.12	0.05	0.03	0.69	0.50	0.36	0.26	23.20	17.00	14.60	13.10	99.40	99.80	100.00	100.00
1,000	0.26	0.20	0.04	0.03	0.50	0.46	0.26	0.19	28.50	24.50	14.90	14.80	100.00	100.00	100.00	100.00

Notes: The DGP is given by (42), where e_{it} are given by (43) and (44). The true parameter values are $\rho = 0.4$, $\beta_1 = 1$, and $\beta_2 = 2$. The power is computed under $H_1 : \rho = 0.38$. The maximum lag of the robust variance estimator is set to be $2\sqrt{T}$. See also the notes to Table 1.

Table 4. Small sample properties of estimators for the slope parameter β_1 ($\beta_1 = 1$, serially correlated and heteroskedastic errors).

	Bias (×100)				RMSE (×100)				Size (×100)				Power (×100)			
$M\backslash T$	20	30	50	100	20	30	50	100	20	30	50	100	20	30	50	100
	Infeasible 2SLS estimator (including factors)															
30	0.10	0.04	−0.02	−0.02	5.35	4.43	3.41	2.38	7.40	7.55	7.30	6.80	21.00	25.75	36.75	59.35
50	−0.23	−0.17	−0.09	−0.12	4.11	3.32	2.53	1.77	6.40	6.45	5.35	5.55	25.55	36.05	51.35	77.65
100	−0.18	−0.07	−0.04	−0.04	2.95	2.37	1.86	1.30	7.05	6.75	7.15	5.70	42.25	60.20	79.90	97.55
500	−0.03	−0.02	−0.01	0.00	1.26	1.04	0.82	0.59	6.00	5.55	6.15	5.90	97.70	99.80	100.00	100.00
1,000	0.02	0.01	0.02	0.00	0.94	0.76	0.60	0.42	7.10	6.70	6.55	6.05	100.00	100.00	100.00	100.00
	2SLS estimator															
30	0.12	0.07	0.05	0.04	5.45	4.49	3.51	2.52	6.65	7.10	8.25	8.45	19.70	26.15	37.15	60.20
50	−0.21	−0.16	−0.06	−0.11	4.17	3.34	2.54	1.79	5.75	6.45	6.05	4.95	24.30	34.00	51.40	78.50
100	−0.14	−0.06	−0.05	−0.04	2.93	2.37	1.86	1.30	5.70	6.85	6.70	5.85	39.85	57.05	78.80	97.50
500	−0.05	−0.04	−0.02	−0.01	1.29	1.05	0.83	0.59	5.30	5.15	5.80	5.95	96.65	99.70	100.00	100.00
1,000	0.02	0.00	0.02	0.00	0.95	0.76	0.60	0.42	6.25	6.35	6.55	5.95	100.00	100.00	100.00	100.00
	B2SLS estimator															
30	0.13	0.07	0.05	0.04	5.45	4.49	3.51	2.51	6.65	7.10	8.30	8.45	19.65	26.20	37.25	60.10
50	−0.20	−0.15	−0.06	−0.11	4.17	3.34	2.54	1.79	5.75	6.50	6.05	5.00	24.40	34.00	51.40	78.55
100	−0.14	−0.06	−0.05	−0.03	2.93	2.37	1.86	1.30	5.75	6.75	6.55	5.90	39.85	57.10	78.85	97.50
500	−0.05	−0.04	−0.02	−0.01	1.29	1.05	0.83	0.59	5.30	5.15	5.80	5.95	96.70	99.65	100.00	100.00
1,000	0.02	0.00	0.02	0.00	0.95	0.76	0.60	0.42	6.35	6.35	6.40	5.95	100.00	100.00	100.00	100.00
	GMM estimator															
30	0.21	0.19	0.17	0.18	5.55	4.55	3.55	2.55	7.80	8.25	8.95	8.75	22.00	28.40	39.35	63.40
50	−0.15	−0.08	0.01	−0.02	4.21	3.34	2.53	1.79	6.40	6.75	6.20	5.40	26.00	35.65	53.40	80.95
100	−0.11	−0.01	0.01	0.02	2.94	2.38	1.86	1.30	6.00	6.70	7.10	5.75	40.90	58.45	79.40	98.05
500	−0.05	−0.03	−0.01	0.00	1.29	1.05	0.82	0.59	5.40	4.95	5.75	5.80	96.75	99.70	100.00	100.00
1,000	0.02	0.01	0.03	0.01	0.95	0.76	0.60	0.42	6.00	6.60	6.85	6.10	100.00	100.00	100.00	100.00
	MLE															
20	0.09	0.02	0.09	0.06	6.94	5.36	4.18	2.95	22.00	16.90	17.55	16.15	34.65	36.45	45.05	62.15
30	0.10	−0.01	−0.05	−0.03	5.61	4.47	3.41	2.35	20.80	18.25	17.10	14.60	38.15	42.60	53.20	74.20
50	−0.16	−0.11	−0.11	−0.12	4.17	3.25	2.43	1.74	18.65	16.15	12.95	13.00	45.10	54.15	68.85	89.95
100	−0.10	−0.06	−0.06	−0.06	2.99	2.32	1.82	1.26	19.85	17.15	16.00	14.10	64.50	77.25	89.60	99.40
500	0.03	0.02	−0.01	−0.00	1.32	1.04	0.79	0.56	19.60	15.40	15.40	14.50	99.50	99.90	100.00	100.00
1,000	0.07	0.01	0.04	0.01	0.96	0.73	0.57	0.40	20.50	17.00	16.60	13.50	100.00	100.00	100.00	100.00

Notes: The DGP is given by (42), where e_{it} are given by (43) and (44). The true parameter values are $\rho = 0.4$, $\beta_1 = 1$, and $\beta_2 = 2$. The power is computed under $H_1 : \beta_1 = 0.95$. The maximum lag of the robust variance estimator is set to be $2\sqrt{T}$. See also the notes to Table 1.

A MSA is defined by the United States Office of Management and Budget as a core area with a relatively high population density (50,000 people or more), including surrounding territory displaying a high level of economic and social integration with the core, as measured by commuting ties. We consider a total of 377 MSAs using the February 2013 delineations, excluding two MSAs in Alaska and two in Hawaii.[34] For the house price data, we use the Freddie Mac House Price Index (FMHPI) at the MSA level covering the period of 1975Q1–2014Q4. The FMHPI is constructed using a repeat-transactions methodology and published by Freddie Mac every quarter. The nominal house prices are deflated by the Consumer Price Index (CPI) for each MSA, and the following analysis is centered on the quarterly rate of changes in real house prices. For the explanatory variables, we are interested in examining the impact of population growth and real per capita income growth on house price growth. See Appendix C for a detailed description of the data sources and transformations.

As a preliminary examination of the data, we conduct the cross-sectional dependence (CD) test developed by Pesaran (2015) on the rate of changes in deseasonalized real house prices. The deseasonalization is performed by regressing the nominal house price changes on seasonal dummies and an intercept for each MSA. The CD statistic turns out to be 1364.110 (with the estimated average of the pairwise correlation coefficient being 0.406), which substantially exceeds the critical value of 1.96 at the 5% level and strongly rejects the null hypothesis of weak cross-sectional dependence. Additionally, we compute the exponent of cross-sectional dependence proposed by Bailey, Kapetanios, and Pesaran (2016) and obtain a value of 1.000 (with a standard error 0.024). The value of the exponent, if it lies within the range $[3/4, 1]$, would suggest that the cross-sectional dependence is fairly strong; lying in $[1/2, 3/4)$, it would imply weak dependence of different degrees. Accordingly, the values of both the CD statistic and the estimated exponent clearly indicate the existence of strong cross-sectional dependence in real house price changes; hence, it is imperative to incorporate common factors into the standard spatial models, which capture only weak cross-sectional dependence.

5.1. The model

Let y_{it} denote the rate of changes in real house prices for area i at time t, which is computed by $y_{it} = \log(P_{it}/\text{CPI}_{it}) - \log(P_{i,t-1}/\text{CPI}_{i,t-1})$, where P_{it} is the house price index and CPI_{it} is the CPI. We consider the following model for house price changes written in stacked form:

$$\mathbf{y}_{.t} = \rho\mathbf{W}\mathbf{y}_{.t} + (\beta_1 + \theta_1\mathbf{W})\%\Delta\ \text{Population}_{.t} + (\beta_2 + \theta_2\mathbf{W})\%\Delta\ \text{Income}_{.t} + \Upsilon\mathbf{d}_t + \mathbf{\Gamma}\mathbf{f}_t + \mathbf{e}_{.t}, \quad (45)$$

for $t = 1, 2, ..., T$, where $\mathbf{y}_{.t} = (y_{1t}, y_{2t}, ..., y_{NT})'$; $\mathbf{d}_t$ denotes an $m_d \times 1$ vector of observed factors that includes quarterly dummies and an intercept; $\mathbf{f}_t$ represents an $m_f \times 1$ vector of unobserved factors; Υ and $\mathbf{\Gamma}$ are corresponding matrices of individual-specific factor loadings; and $\mathbf{e}_{.t}$ is a vector of idiosyncratic errors. It should be noted that this model accommodates individual fixed effects by including a constant term in $\mathbf{d}_t$ and letting its loadings be heterogeneous. $\%\Delta\text{Population}_{.t}$ represents an $N \times 1$ vector of percentage changes in population at time t, and $\%\Delta\text{Income}_{.t}$ denotes a vector of percentage changes in real per capita income. Both variables are calculated as first differences of natural logarithms. $\mathbf{W}$ is the spatial weights matrix. For generality, model (45) also allows for spatial lags of the explanatory variables, namely $\mathbf{W}\%\Delta\text{Population}_{.t}$ and $\mathbf{W}\%\Delta\text{Income}_{.t}$, which are often referred to as Durbin terms in the literature and capture the interaction effects of exogenous variables.

[34]The Office of Management and Budget periodically revises the MSA delineations to reflect the changes in population counts and commuting patterns. There are 381 MSAs in the US as of February 2013. The terms "area" and "MSA" are used interchangeably in the following discussions.

Table 5. GMM estimation results of model (45).

%ΔHouse price	(1)	(2)	(3)	(4)	(5)	(6)
ρ [$\mathbf{W} \times$ %ΔHouse price]	0.730***	0.643***	0.732***	0.648***	0.731***	0.648***
	(0.004)	(0.005)	(0.004)	(0.005)	(0.004)	(0.005)
β_1 [%ΔPopulation]	0.380***	0.366***	0.383***	0.432***	0.369***	0.417***
	(0.035)	(0.040)	(0.037)	(0.048)	(0.036)	(0.048)
β_2 [%ΔIncome per capita]	0.099***	0.093***	0.106***	0.096***	0.111***	0.094***
	(0.007)	(0.007)	(0.007)	(0.008)	(0.008)	(0.008)
θ_1 [$\mathbf{W} \times$ %ΔPopulation]			0.078**	0.063*	0.063**	0.069*
			(0.031)	(0.036)	(0.031)	(0.037)
θ_2 [$\mathbf{W}\times$ %ΔIncome per capita]					−0.006	0.019
					(0.010)	(0.012)
Regional unobserved factors	No	Yes	No	Yes	No	Yes
National unobserved factors	Yes	Yes	Yes	Yes	Yes	Yes
MSA FE and seasonal dummies	Yes	Yes	Yes	Yes	Yes	Yes
Residuals						
CD test statistic	−4.946	−6.532	−4.927	−6.385	−5.111	−6.365
Exponent of cross-section dependence	0.734***	0.674***	0.743***	0.690***	0.734***	0.652***
	(0.031)	(0.019)	(0.030)	(0.019)	(0.027)	(0.019)
$\bar{R}^2$	0.808	0.837	0.813	0.844	0.817	0.847
Observations	$N = 377, T = 159$					

Notes: Dependent variable is the rate of changes in real house prices. The explanatory variables are population growth rate and real per capita income growth rate, as well as possibly their spatial lags. MSAs are classified into eight BEA Regions. All estimations consider national unobserved factors and include MSA fixed effects (FE) and quarterly dummies. Factor estimates are not reported in order to save space. The spatial weights matrix is $\mathbf{W} = \mathbf{W}_{100}$. The efficient GMM estimates are obtained by (40), using $\mathbf{P}_1 = \mathbf{W}$ and $\mathbf{P}_2 = \mathbf{W}^2 - Diag(\mathbf{W}^2)$ in the quadratic moments and $\mathbf{Q}_{.t}^{[2]} = (\mathbf{X}_{.t}, \mathbf{W}\mathbf{X}_{.t}, \mathbf{W}^2\mathbf{X}_{.t})$ as IVs in the linear moments. Standard errors are in parentheses. The standard errors for the slope estimates are heteroskedasticity and autocorrelation consistent with the maximum lag length set to $2\lfloor T^{1/2} \rfloor$. *, **, *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

When it comes to the specification of $\mathbf{W}$, it is common practice to adopt distance- or contiguity- based weighting scheme in the studies of spatial dependence in housing markets. We will follow this tradition first and then explore other possibilities in the subsequent section. In particular, we assume that the contiguity relations are determined by radial distance and define "neighbors" of an MSA as those units located within a threshold distance d (miles). The weights of neighbors take a value of one, and the weights of non-neighbors take a value of zero. Then, $\mathbf{W}$ is row-standardized so that the weights across each row sums to unity. The spatial weights matrix constructed in this way is denoted by $\mathbf{W}_d$. Our analysis takes $d = 100$ miles as a point of departure and examines potential dependencies within commuting and transport distances around an MSA.

The parameters of interest are $\boldsymbol{\delta} = (\rho, \boldsymbol{\beta}', \boldsymbol{\theta}')'$, where $\boldsymbol{\beta} = (\beta_1, \beta_2)'$ and $\boldsymbol{\theta} = (\theta_1, \theta_2)'$. In what follows, we will focus on the efficient GMM estimator of $\boldsymbol{\delta}$ defined by (40).[35] Specifically, the estimation is implemented by using $\mathbf{P}_1 = \mathbf{W}$ and $\mathbf{P}_2 = \mathbf{W}^2 - Diag(\mathbf{W}^2)$ in the quadratic moments and $\mathbf{Q}_{.t}^{[2]} = (\mathbf{X}_{.t}, \mathbf{W}\mathbf{X}_{.t}, \mathbf{W}^2\mathbf{X}_{.t})$ as instruments in the linear moments, where $\mathbf{X}_{.t} = (\Delta\text{Population}_{.t}, \%\Delta\text{Income}_{.t})$, for $t = 1, 2, ..., T$.

Table 5 summarizes the estimation results of model (45) based on $\mathbf{W} = \mathbf{W}_{100}$. Findings using other specifications of $\mathbf{W}$ will be discussed later. In column (1), the Durbin terms are excluded and the unobserved factors are proxied by cross-sectional averages of both regressand and regressors across all MSAs.[36] The estimated spatial coefficient is positive and highly significant, with a value of 0.730 (with a standard error of 0.004). Higher population and income growth are found

[35]The 2SLS estimates are omitted to save space, since they are very close to the GMM estimates with slightly larger standard errors.

[36]In the empirical analysis, $\bar{y}_t^* = N^{-1}\sum_{i=1}^{N} y_{it}^*$ is also included as factor proxies since it may potentially improve the small sample properties of the estimator. Nonetheless, whether $\bar{y}_t^*$ is included makes little difference to the results since $\bar{y}_t^*$ and $\bar{y}_t$ are highly correlated for most of the $\mathbf{W}$ matrices we considered.

to increase house price growth, as anticipated. We then include the Durbin terms, and we add to the list of factor proxies the cross-sectional averages of $\mathbf{X}_{.t}^{*}$ across all MSAs, where $\mathbf{X}_{.t}^{*}=\mathbf{W}\mathbf{X}_{.t}$. As can be seen from columns (3) and (5), population growth displays a positive and significant spatial interaction effect, but real income growth does not. Overall, the estimates of ρ and $\boldsymbol{\beta}$ are very close across columns (1), (3), and (5). The CD statistics on the residuals of these specifications range from -5.11 to -4.93, which are substantially reduced from the previous test statistic, 1364.110, of the house price growth series itself. The exponents of cross-section dependence of the residuals, however, are about 0.73–0.74, which suggests that a moderate degree of cross-section dependence may be unaccounted for. Therefore, we will next consider local (regional) unobserved factors in addition to global (national) factors, and we will investigate if strong dependence can be more effectively eliminated.[37]

Suppose now that all MSAs are classified into R regions. The model can still be represented by (45), but the observations are now ordered by regions. In specific, the $N\times 1$ vector of house prices changes, $\mathbf{y}_{.t}$, can be written as $\mathbf{y}_{.t}=(\mathbf{y}_{.1t}',\mathbf{y}_{.2t}',...,\mathbf{y}_{.Rt}')'$, where $\mathbf{y}_{.rt}=(y_{1rt},y_{2rt},...,y_{N_rrt})'$ is an $N_r\times 1$ vector of observations for the rth region, for $r=1,2,...,R$. N_r is the number of MSAs in region r, and clearly, we have $\sum_{r=1}^{R}N_r=N$. Observations on independent variables and spatial weights are also sorted accordingly. Note that the latent factors, $\mathbf{f}_t$, are now assumed to have a hierarchical structure, namely, $\mathbf{f}_t=(\mathbf{f}_{g,t}',\mathbf{f}_{l,t}')'$, where $\mathbf{f}_{g,t}$ denotes an $m_g\times 1$ vector of global factors, $\mathbf{f}_{l,t}$ denotes an $m_l\times 1$ vector of local factors, and $m_g+m_l=m_f$. The associated factor loadings are partitioned as $\boldsymbol{\Gamma}=(\boldsymbol{\Gamma}_g,\boldsymbol{\Gamma}_l)$, where $\boldsymbol{\Gamma}_g=(\boldsymbol{\gamma}_{g,1},\boldsymbol{\gamma}_{g,2},...,\boldsymbol{\gamma}_{g,N})'$ is an $N\times m_g$ matrix of loadings for national factors, and $\boldsymbol{\Gamma}_l=(\boldsymbol{\Gamma}_{l,1}',\boldsymbol{\Gamma}_{l,2}',...,\boldsymbol{\Gamma}_{l,R}')'$ is an $N\times m_l$ matrix, with $\boldsymbol{\Gamma}_{l,r}=(\boldsymbol{\gamma}_{l,1r},\boldsymbol{\gamma}_{l,2r},...,\boldsymbol{\gamma}_{l,N_rr})'$ being the $N_r\times m_l$ factor loadings for the rth region, $r=1,2,...,R$. The proposed GMM estimation procedure can easily accommodate regional unobservable factors by replacing them with cross-sectional averages of observations on both dependent and individual-specific independent variables for that region.

Table 5 columns (2), (4), and (6) report the estimation results when both regional and national latent factors are taken into account. We group all 377 MSAs into $R=8$ Regions based on the geographical classification by the Bureau of Economic Analysis (BEA).[38] The model's goodness of fit is measured by $\bar{R}^2$.[39] Compared with the earlier outcomes that did not assume regional factors, both the estimated spatial coefficients and the exponents of cross-section dependence of the residuals decline, suggesting that regional common shocks contribute to the strong cross-sectional dependence in house price changes in the US and that the strength of spatial connections will be overestimated if strong dependence is not effectively eliminated. In addition, after the inclusion of regional factors, the spatial interaction effect of population growth is no longer significant at the 5% level; the spatial interaction effect of income growth remains insignificant. According to the above comparisons, we conclude that column (2) provides the best estimation results among all the specifications in Table 5, which points to a significant neighborhood effect in house price changes, with an estimated spatial coefficient of 0.643 (0.005).[40]

Care must be taken when interpreting the estimates of $\boldsymbol{\beta}$ and $\boldsymbol{\theta}$ in model (45), as they do not directly signify the marginal effects of the independent variables on house price variations. An important feature of SAR models is that a change in an explanatory variable of a unit will affect

[37]Bailey, Holly, and Pesaran (2016) also consider regional effects, but the authors do not show the impact of eliminating regional factors to the estimated intensity of spatial dependence.

[38]The eight BEA Regions are New England, Mideast, Great Lakes, Plains, Southeast, Southwest, Rocky Mountain, and Far West Regions.

[39]$\bar{R}^2=1-\hat{\sigma}_{\text{res}}^2/\hat{\sigma}_{\text{tot}}^2$, where $\hat{\sigma}_{\text{res}}^2=[N(T-k_{cs}-m_d)-k_z]^{-1}\sum_{i=1}^{N}(\mathbf{y}_{i.}-\mathbf{Z}_{i.}\hat{\boldsymbol{\delta}})'\bar{\mathbf{M}}(\mathbf{y}_{i.}-\mathbf{Z}_{i.}\hat{\boldsymbol{\delta}})$, $\hat{\sigma}_{\text{tot}}^2=[N(T-1)]^{-1}\sum_{i=1}^{N}\sum_{t=1}^{T}(y_{it}-\bar{y}_{i.})^2$, $\bar{y}_{i.}=T^{-1}\sum_{t=1}^{T}y_{it}$, $\mathbf{y}_{i.}=(y_{i1},y_{i2},...,y_{iT})'$, m_d is the number of observed factors, and $\bar{\mathbf{M}}$ represents the de-factoring matrix of $T\times k_{cs}$ dimension. In specific, $m_d=4$ since the observed factors consist of quarterly dummies and an intercept. The values of k_z and k_{cs} vary with detailed model specifications, that is, whether Durbin terms are included and regional factors are considered. This measure of model fit in the presence of unobserved factors is in accordance with the suggestion by Holly et al. (2010, p. 164).

[40]Standard error in parenthesis.

not only the dependent variable of that unit itself but also the dependent variables of other cross-section units. The former is known as the direct effect, the latter as indirect effect or spillover effect. Also notice that both effects in general vary across cross-section units. Therefore, to find out the marginal effects of population and income growth on house price changes, we calculate the summary measures of direct and indirect effects proposed by LeSage and Pace (2009). The average direct effect of the kth explanatory variable ($k = 1, 2$) is given by the average of the diagonal elements of $\boldsymbol{\Pi}_k$, where

$$\boldsymbol{\Pi}_k = (\mathbf{I}_N - \rho\mathbf{W})^{-1}(\beta_k\mathbf{I}_N + \theta_k\mathbf{W}), \tag{46}$$

and the average indirect effect is represented by the average row sum of the nondiagonal elements of $\boldsymbol{\Pi}_k$. It can be seen from (46) that imposing $\boldsymbol{\theta} = \mathbf{0}$ implies that the ratio of direct to indirect effects is the same for every explanatory variable, which might be too restrictive; hence, model (45) takes into account the Durbin terms. To test if the direct and spillover effects are significant, we compute the standard errors by simulation, due to the complex formula for the effects in terms of the parameters.[41]

Table 6 shows the estimated average direct and spillover effects of population and income growth on house price growth based on the estimates in Table 5. The average total effect is the sum of average direct and indirect effects. When both national and regional unobserved factors are considered, the specification in column (2) of Table 5 outperforms its counterpart. When only national factors are taken into account, the preferred specification is given by column (3). We compute the effects of the explanatory variables based on these estimates, respectively. It is not surprising to see from Table 6 that the estimated indirect effects are much higher if regional factors are neglected, as there is a relatively stronger degree of cross-sectional dependence in house prices left uncontrolled for. Both population growth and per capita income growth are found to exert both positive and significant direct and indirect impact on house price changes. Specifically, using the estimates produced assuming a hierarchical factor structure, on average a one percentage point increase in population growth in an MSA is predicted to lead to a 0.43 percentage point increase in house price growth in the MSA itself, and a 0.57 percentage point increase in house price growth in its neighboring MSAs, while holding other covariates fixed. In comparison, a one percentage point increase in income growth has much smaller direct and indirect effects on house price growth, which are estimated to be 0.11 and 0.15 percentage points, respectively. The spillover effect of population growth to neighboring MSAs appears to be slightly higher than its direct effect, while both effects of income growth are of similar magnitude.

5.2. Different spatial weights matrix specifications

We now turn to inspecting the robustness of our findings to various specifications of the spatial weights matrix. Three types of weights are considered, which are constructed based on distance, migration flows, and pairwise correlations, respectively. In all of the following analysis, we will control for unobserved factors at both national and regional levels, as the earlier discussion has revealed the importance of both effects on the cross-sectional dependence in house price changes.

We start with comparing the estimation results of model (45) using different radial distance matrices, $\mathbf{W}_d$. In specific, we consider three threshold values, $d = 75$, 100, and 125 miles. The estimation results are presented in columns (1) to (3) of Table 7, respectively. Overall, the estimates are found to be very stable as the cutoff distance varies. The estimated strength of spatial dependence rises slightly from 0.573(0.005) to 0.693(0.005) as the neighborhood boundary expands from 75 to 125 miles. This change is reasonable because more units are considered as neighbors of an MSA and their influences are taken into account. The average number of neighbors per

[41]See LeSage and Pace (2009) and Section 2.7 of Elhorst (2014) for detailed discussions on the computation.

Table 6. Average direct and indirect effects of population and income growth on house price changes.

	Direct	Indirect	Total
Considering both national and regional factors			
%ΔPopulation	0.431***	0.571***	1.002***
	(0.047)	(0.063)	(0.110)
%ΔIncome per capita	0.110***	0.146***	0.256***
	(0.009)	(0.012)	(0.020)
Considering national factors only			
%ΔPopulation	0.518***	1.153***	1.672***
	(0.046)	(0.112)	(0.149)
%ΔIncome per capita	0.135***	0.249***	0.384***
	(0.009)	(0.017)	(0.026)

Notes: The effects of explanatory variables, taking both national and regional factors into account, are computed based on the estimates in column (2) of Table 5. When regional factors are neglected, the effects are computed using the estimates in column (3) of Table 5. Bootstrapped standard errors based on 1,000 replications are in parentheses. *** indicates statistical significance at the 1% level. See also the notes to Table 5.

MSA is 3.31 when $\mathbf{W} = \mathbf{W}_{75}$, as compared to 8.65 when $\mathbf{W} = \mathbf{W}_{125}$. In addition, the estimated coefficients of population and income growth remain in relatively narrow ranges as d changes. Both variables are highly significant and of reasonable magnitude.

Since many economic and demographic factors apart from geographical proximity may contribute to the cross-sectional dependence in house prices across MSAs, it is interesting to consider spatial weights based on other measures of closeness. In particular, the MSA-to-MSA migration flows are important indicators of the strength of interconnections. We construct a migration weights matrix, denoted by $\mathbf{W}_m$, of which the $(i,j)^{th}$ element represents the share of movers from area j to area i of the total number of movers to area i. We do not consider non-movers or migration flows from/to non-MSAs. Notice that $\mathbf{W}_m$ is an asymmetric matrix in which the immigration flow to each MSA is normalized to unity. The data on inter-MSA migration flows were introduced as part of the American Community Survey (ACS) dataset since 2009; therefore, $\mathbf{W}_m$ is constructed using the migration data from the 2010 to 2014 ACS 5-year estimates. After dropping the estimates with high margin of errors, each MSA ends up having an average of 4.46 "neighbors."[42] Since the most dominant migration ties are likely to be stable over a long period of time, the time invariability of $\mathbf{W}_m$ does not give cause for concern.

The estimation results of model (45) using $\mathbf{W}_m$ are reported in column (4) of Table 7. Not surprisingly, we find strong evidence of spatial dependence based on migration relations. The estimated spatial parameter is significantly positive, slightly higher than the estimates using distance-based weights. The estimated coefficients of population and income growth are very close to the previous results, and both are significantly different from zero. Both residual diagnostics and the value of $\bar{R}^2$ indicate that the model is a good fit. The similarities between the results using distance and migration weights are quite striking, given that around 65% of the migration flows occur between MSAs located at most 100 miles apart.[43]

The third type of spatial weights matrix we consider is constructed by a data-driven approach that detects significant bilateral relations using house price series itself. In essence, this approach equates significant pairwise correlations with significant connections. Bailey, Holly, and Pesaran (2016) suggest filtering out strong cross-section dependence from a series first, then applying a regularization or thresholding method to create sparse weights matrices. We follow this idea and construct weights matrices based on significantly positive and negative pair-wise correlations of de-factored house price changes, which are denoted by $\hat{\mathbf{W}}^{+}$ and $\hat{\mathbf{W}}^{-}$, respectively. Specifically,

[42]Details on the construction of $\mathbf{W}_m$ and its characterization are given in Appendix C and the Supporting Information.
[43]See Figure S.1 in the Supporting Information for the distribution of distance between the area of origin and the area of destination.

Table 7. GMM estimation results of model (45) using different spatial weights matrices.

	Spatial weights matrix				
	Distance			Migration	Pairwise correlations
	$\mathbf{W}_{75}$	$\mathbf{W}_{100}$	$\mathbf{W}_{125}$	$\mathbf{W}_m$	$\hat{\mathbf{W}}^+, \hat{\mathbf{W}}^-$
%ΔHouse price	(1)	(2)	(3)	(4)	(5)
ρ [$\mathbf{W} \times$ %ΔHouse price]	0.573***	0.643***	0.693***	0.772***	
	(0.005)	(0.005)	(0.005)	(0.005)	
ρ^+ [$\hat{\mathbf{W}}^+ \times$ %ΔHouse price]					0.715***
					(0.005)
ρ^- [$\hat{\mathbf{W}}^- \times$ %ΔHouse price]					−0.308***
					(0.005)
β_1 [%ΔPopulation]	0.432***	0.366***	0.294***	0.230***	0.147***
	(0.052)	(0.040)	(0.036)	(0.031)	(0.023)
β_2 [%ΔIncome per capita]	0.099***	0.093***	0.089***	0.075***	0.049***
	(0.008)	(0.007)	(0.007)	(0.007)	(0.005)
Natl. and Rgnl. unobserved factors	Yes	Yes	Yes	Yes	Yes
MSA FE and seasonal dummies	Yes	Yes	Yes	Yes	Yes
Residuals					
CD test statistic	−6.678	−6.532	−7.127	−3.114	−6.846
Exponent of cross-section dependence	0.668***	0.674***	0.624***	0.728***	0.631***
	(0.023)	(0.019)	(0.017)	(0.021)	(0.014)
Avg. no. neighbors	3.31	5.73	8.65	4.46	11.01 [$\hat{\mathbf{W}}^+$], 8.02 [$\hat{\mathbf{W}}^-$]
$\bar{R}^2$	0.833	0.837	0.833	0.840	0.908
Observations	$N = 377, T = 159$				

Notes: All estimations consider both national and regional (Natl. and Rgnl.) unobserved factors and also include MSA fixed effects (FE) and quarterly dummies. Factor estimates are not reported to save space. $\mathbf{W}_d$ denotes radial distance weights matrix with threshold distance d, where $d = 75$, 100, and 125 miles. $\mathbf{W}_m$ denotes weights matrix constructed from MSA-to-MSA migration flows. $\hat{\mathbf{W}}^+$ and $\hat{\mathbf{W}}^-$ denote weights matrices constructed from significant positive and negative pairwise correlations of de-factored house price changes, respectively. *** indicates statistical significance at the 1% level. See also the notes to Table 5.

the de-factoring process is conducted by regressing the house price growth rate for each MSA on an intercept, quarterly dummies, and cross-sectional averages of the dependent and explanatory variables at both national and regional levels. Then, significant connections are identified by applying the multiple testing procedure developed by Bailey, Pesaran, and Smith (2019) to the sample correlation matrix of the first-step residuals at the 5% significance level. If the corresponding correlation coefficient is positively significant, the element of $\hat{\mathbf{W}}^+$ is set to one, otherwise to zero. $\hat{\mathbf{W}}^-$ is constructed similarly but based on significantly negative correlations. $\hat{\mathbf{W}}^+$ and $\hat{\mathbf{W}}^-$ are then row-standardized so that each row sums to one.[44]

With the correlation-based weights matrices, we are able to distinguish between the intensity of positive and negative spatial connections. Let us now consider the following model,

$$\mathbf{y}_{.t} = \rho^+ \hat{\mathbf{W}}^+ \mathbf{y}_{.t} + \rho^- \hat{\mathbf{W}}^- \mathbf{y}_{.t} + \beta_1 \%\Delta\text{Population}_{.t} + \beta_2 \%\Delta\text{Income}_{.t} + \Upsilon \mathbf{d}_t + \Gamma \mathbf{f}_t + \mathbf{e}_{.t}, \tag{47}$$

where, as before, $\mathbf{d}_t$ includes an intercept and quarterly dummies, and $\mathbf{f}_t$ contains national and regional unobserved factors.[45]

Column (5) of Table 7 presents the estimation results of model (47). The estimated ρ^+ and ρ^- have the correct sign, and both are highly significant. The magnitude of the positive spatial effect is notably greater than that of the negative effect, with $\hat{\rho}^+$ amounting to $0.715(0.005)$ and $\hat{\rho}^-$ being $-0.308(0.005)$. The coefficients of population and income growth are again found to be positive and significant, with slightly smaller magnitude than those obtained using distance and migration weights matrices. The CD statistic is low, and the cross-section exponent is close to the

[44]See Appendix C and the Supporting Information for a more detailed characterization and comparison of different spatial weights matrices.

[45]We have also considered the Durbin terms, which are found to be insignificant.

borderline case of 0.5, suggesting that only weak dependence is left in the residuals. Overall, the model fits the data very well, as shown by the value of $\bar{R}^2$.

6. Concluding remarks

This article considers panel data models in the presence of two sources of cross-sectional dependence: endogenous spatial interactions and common effects and develops 2SLS/GMM estimations for the joint model. The estimation approaches replace the unobserved common factors with cross-sectional averages and utilize instrumental variables and quadratic moment conditions in order to cope with the endogenous spatial effects. The proposed estimators are shown to be consistent as long as N is large, irrespective of the size of T. The asymptotic distributions of these estimators are Gaussian and free of asymptotic bias, provided that T is of a smaller order of magnitude than N, as both N and T tend to infinity jointly. Compared with the maximum likelihood approach, the number of latent factors need not be estimated, and general forms of serial correlation in the disturbances are permitted. A wide range of Monte Carlo exercises lend further support to these theoretical results.

A detailed empirical application to real house price changes reveals that significant spatial dependence exists across MSAs in the US, and it demonstrates the importance of adequately removing common effects when analyzing the strength of spatial interconnections. The study also identifies significant effects of population and income growth on house price growth. Besides geographical proximity, we also consider spatial weights based on migration flows and on pairwise correlations of de-factored house price changes. The main findings remain valid under the different measures of connections. These empirical results highlight the need to consider the spatial spillover effects in housing markets when making policy and business decisions.

In accord with the most recent literature, the 2SLS/GMM estimators are asymptotically biased if T/N approaches a finite positive constant as both N and T tend to infinity. Therefore, a natural next step is to investigate bias correction procedures for this case. An interesting extension of the current model is to incorporate slope heterogeneity, which is especially relevant for applications covering different countries, regions, or industries. The current article is also related to the recent study by Pesaran and Yang (2020a), who consider networks in the presence of dominant units and common factors. In such scenarios, the spatial weights matrix may have unbounded column sums, and the identification and estimation problems are of practical importance and worth further investigation.

Appendix A: Notations

This appendix summarizes some frequently used notations and variable definitions in the main article and Appendix B.

K is used generically for a finite positive constant. For an $N \times N$ real matrix $\mathbf{A} = (a_{ij})$, $||\mathbf{A}|| = \sqrt{tr(\mathbf{A}\mathbf{A}')}$, $||\mathbf{A}||_\infty = \max_{1\le i\le N} \sum_{j=1}^{N} |a_{ij,N}|$, and $||\mathbf{A}||_1 = \max_{1\le j\le N} \sum_{i=1}^{N} |a_{ij}|$ denote the Frobenius norm, the maximum row sum norm, and maximum column sum norm of matrix $\mathbf{A}$, respectively. We say that the row (column) sums of a (sequence of) matrix $\mathbf{A}$ are uniformly bounded in absolute value, or $\mathbf{A}$ has bounded row (column) norm for short, if there exists a constant K such that $||\mathbf{A}||_\infty < K$ ($||\mathbf{A}||_1 < K$) for all N. $Diag(\mathbf{A}) = Diag(a_{11}, a_{22}, ..., a_{NN})$ represents an $N \times N$ diagonal matrix formed with the diagonal entries of $\mathbf{A}$, whereas $diag(\mathbf{A}) = (a_{11}, a_{22}, ..., a_{NN})'$ denotes an $N \times 1$ vector. $tr(\mathbf{A})$ denotes the trace of $\mathbf{A}$, and $\text{vec}(\mathbf{A})$ denotes the column vector obtained by stacking the columns of $\mathbf{A}$. $\odot$ and $\otimes$ are the Hadamard product and Kronecker product, respectively.

Let $\{x_N\}_{N=1}^{\infty}$ be any real sequence and $\{y_N\}_{N=1}^{\infty}$ be a sequence of positive real numbers; we write $x_N = O(y_N)$ if there exists a positive finite constant K such that $|x_N| \le Ky_N$ for all N, and $x_N = o(y_N)$ if $x_N/y_N \to 0$ as $N \to \infty$. $O_p(.)$ and $o_p(.)$ are the equivalent stochastic orders in probability. $\sim$ denotes asymptotic equivalence. $(N,T) \xrightarrow{j} \infty$ denotes joint convergence of N and T. $\boldsymbol{\tau}_N$ is an $N \times 1$ vector of ones.

$\boldsymbol{\delta} = (\rho, \boldsymbol{\beta}')'$; their true values are $\boldsymbol{\delta}_0 = (\rho_0, \boldsymbol{\beta}_0')'$.

$\mathbf{S}(\rho) = \mathbf{I}_N - \rho\mathbf{W}$, for any possible value of ρ in the parameter space, and $\mathbf{S}_0 = \mathbf{S}(\rho_0) = \mathbf{I}_N - \rho_0\mathbf{W}$. $\mathbf{G}(\rho) = \mathbf{W}\mathbf{S}^{-1}(\rho)$, $\mathbf{G}_0 = \mathbf{G}(\rho_0) = \mathbf{W}\mathbf{S}_0^{-1}$, and $\mathbf{G}_0^b = \mathbf{I}_T \otimes \mathbf{G}_0$.

$\mathbf{X}_{.t} = (\mathbf{x}_{1t}, \mathbf{x}_{2t}, ..., \mathbf{x}_{Nt})'$ is an $N \times k$ matrix of observations at time t, $\mathbf{X}_{i.} = (\mathbf{x}_{1.}, \mathbf{x}_{2.}, ..., \mathbf{x}_{N.})'$ is a $T \times k$ matrix of observations for unit i, and $\mathbf{X} = (\mathbf{X}'_{.1}, \mathbf{X}'_{.2}, ..., \mathbf{X}'_{.T})'$ is an $NT \times k$ matrix of stacked cross-sections. $\{\mathbf{y}_{.t}, \mathbf{y}_{i.}, \mathbf{Y}\}$, $\{\mathbf{e}_{.t}, \mathbf{e}_{i.}, \mathbf{e}\}$, $\{\mathbf{V}_{.t}, \mathbf{V}_{i.}, \mathbf{V}\}$, $\{\mathbf{Q}_{.t}, \mathbf{Q}_{i.}, \mathbf{Q}\}$, and so forth are defined similarly.

$y_{it}^* = \sum_{j=1}^N w_{ij} y_{jt}$, $\mathbf{y}_{.t}^* = \mathbf{W}\mathbf{y}_{.t} = (y_{1t}^*, y_{2t}^*, ..., y_{Nt}^*)'$, and $\mathbf{Y}^* = (\mathbf{I}_T \otimes \mathbf{W})\mathbf{Y} = (\mathbf{y}_{.1}^{*\prime}, \mathbf{y}_{.2}^{*\prime}, ..., \mathbf{y}_{.T}^{*\prime})'$.

$\mathbf{L} = (\mathbf{Y}^*, \mathbf{X})$ and $\mathbf{L}_0 = (\mathbf{G}_0^b\mathbf{X}\boldsymbol{\beta}_0, \mathbf{X})$.

$\mathbf{z}_{it} = (y_{it}, \mathbf{x}'_{it})'$, $\mathbf{z}_{.t} = (\mathbf{z}'_{1t}, \mathbf{z}'_{2t}, ..., \mathbf{z}'_{Nt})'$ is an $N(k+1) \times 1$ vector, and $\mathbf{Z} = (\mathbf{z}_{.1}, \mathbf{z}_{.2}, ..., \mathbf{z}_{.T})'$.

$\mathbf{u}_{it} = (e_{it}, \mathbf{v}'_{it})'$. $\boldsymbol{\epsilon}_{.t} = \boldsymbol{\Delta}^{-1}\mathbf{u}_{.t} = (\boldsymbol{\epsilon}_{1t}, \boldsymbol{\epsilon}_{2t}, ..., \boldsymbol{\epsilon}_{Nt})'$ is the transformed composite error, where $\boldsymbol{\Delta}^{-1}$ is given by (4) in the main article. $\bar{\boldsymbol{\epsilon}}_{.t} = \boldsymbol{\Theta}_a\boldsymbol{\epsilon}_{.t} = N^{-1}\sum_{i=1}^N \boldsymbol{\epsilon}_{it}$.

$\mathbf{F} = (\mathbf{f}_1, \mathbf{f}_2, ..., \mathbf{f}_T)'$ is a $T \times m$ matrix of factors and $\boldsymbol{\Gamma} = (\boldsymbol{\gamma}_1, \boldsymbol{\gamma}_2, ..., \boldsymbol{\gamma}_N)'$ is an $N \times m$ matrix of factor loadings.

$\mathbf{M}_f = \mathbf{I}_T - \mathbf{F}(\mathbf{F}'\mathbf{F})^-\mathbf{F}'$ and $\mathbf{M}_f^b = \mathbf{M}_f \otimes \mathbf{I}_N$ are infeasible de-factoring matrices based on unknown $\mathbf{F}$.

$\bar{\mathbf{M}} = \mathbf{I}_T - \bar{\mathbf{Z}}(\bar{\mathbf{Z}}'\bar{\mathbf{Z}})^-\bar{\mathbf{Z}}'$ and $\bar{\mathbf{M}}^b = \bar{\mathbf{M}} \otimes \mathbf{I}_N$ are the de-factoring matrices based on $\bar{\mathbf{Z}} = (\bar{\mathbf{z}}_{.1}, \bar{\mathbf{z}}_{.2}, ..., \bar{\mathbf{z}}_{.T})'$, where $\bar{\mathbf{z}}_{.t} = (\bar{y}_{.t}, \bar{\mathbf{x}}'_{.t})'$, $\bar{y}_{.t} = N^{-1}\sum_{i=1}^N y_{it}$ and $\bar{\mathbf{x}}_{.t} = N^{-1}\sum_{i=1}^N \mathbf{x}_{it}$.

Appendix B: Proofs of Main Theorems

The following proofs make use of the lemmas provided in the Supporting Information.

Proof of Theorem 1. We begin by showing that $\hat{\boldsymbol{\delta}}_{2sls}$ is consistent for $\boldsymbol{\delta}_0$ as $N \to \infty$, for T fixed or $T \to \infty$. Note that

$$\hat{\boldsymbol{\delta}}_{2sls} - \boldsymbol{\delta}_0 = \left(\mathbf{L}'\mathbf{P}_Q\mathbf{L}\right)^{-1}\mathbf{L}'\mathbf{P}_Q[(\mathbf{I}_T \otimes \boldsymbol{\Gamma}_0)\mathbf{f} + \mathbf{e}], \tag{B.1}$$

where $\mathbf{e} = (\mathbf{e}'_{.1}, \mathbf{e}'_{.2}, ..., \mathbf{e}'_{.T})'$ is $NT \times 1$, $\mathbf{f} = (\mathbf{f}'_1, \mathbf{f}'_2, ..., \mathbf{f}'_T)'$ is $mT \times 1$, $\boldsymbol{\Gamma}_0 = (\boldsymbol{\gamma}_{1,0}, \boldsymbol{\gamma}_{2,0}, ..., \boldsymbol{\gamma}_{N,0})'$ is the $N \times m$ matrix of factor loadings at the true population values. (B.1) can be rewritten as

$$\begin{aligned}\sqrt{NT}(\hat{\boldsymbol{\delta}}_{2sls} - \boldsymbol{\delta}_0) = &\left[\frac{1}{NT}\mathbf{L}'\mathbf{M}^b\mathbf{Q}\left(\frac{1}{NT}\mathbf{Q}'\mathbf{M}^b\mathbf{Q}\right)^{-1}\frac{1}{NT}\mathbf{Q}'\mathbf{M}^b\mathbf{L}\right]^{-1} \\ &\times \left\{\frac{1}{NT}\mathbf{L}'\mathbf{M}^b\mathbf{Q}\left(\frac{1}{NT}\mathbf{Q}'\mathbf{M}^b\mathbf{Q}\right)^{-1}\frac{1}{\sqrt{NT}}\mathbf{Q}'\mathbf{M}^b[(\mathbf{I}_T \otimes \boldsymbol{\Gamma}_0)\mathbf{f} + \mathbf{e}]\right\}.\end{aligned} \tag{B.2}$$

By Lemma S.7, we have

$$\frac{1}{NT}\mathbf{Q}'\mathbf{M}^b\mathbf{Q} = \frac{1}{NT}\mathbf{Q}'\mathbf{M}_f^b\mathbf{Q} + O_p\left(\frac{1}{N}\right) + O_p\left(\frac{1}{\sqrt{NT}}\right), \tag{B.3}$$

$$\frac{1}{NT}\mathbf{Q}'\mathbf{M}^b\mathbf{L} = \frac{1}{NT}\mathbf{Q}'\mathbf{M}_f^b\mathbf{L}_0 + O_p\left(\frac{1}{N}\right) + O_p\left(\frac{1}{\sqrt{NT}}\right), \tag{B.4}$$

where $\mathbf{L}_0$ is defined by (13) in the main article. It follows that

$$\underset{N\to\infty}{p\lim}\frac{1}{NT}\mathbf{L}'\mathbf{P}_Q\mathbf{L} = \underset{N\to\infty}{p\lim}\frac{1}{NT}\mathbf{L}_0'\mathbf{P}_{Q,f}\mathbf{L}_0,$$

where $\mathbf{P}_{Q,f}$ is given by (20) in the main article, and the limit exists and is nonsingular under Assumption 8. Furthermore, by Lemma S.7 we have $\underset{N\to\infty}{p\lim}(NT)^{-1}\mathbf{Q}'\mathbf{M}_f^b[(\mathbf{I}_T \otimes \boldsymbol{\Gamma}_0)\mathbf{f} + \mathbf{e}] = \mathbf{0}$. Therefore, $\hat{\boldsymbol{\delta}}_{2sls}$ is consistent for $\boldsymbol{\delta}_0$, as $N \to \infty$.

To establish the asymptotic distribution, we need to focus on $(NT)^{-1/2}\mathbf{Q}'\mathbf{M}^b[(\mathbf{I}_T \otimes \boldsymbol{\Gamma})\mathbf{f} + \mathbf{e}]$. Note that any column of $\mathbf{Q}$ can be expressed generically as $\mathbf{Q}_c = vec(\mathbf{W}^r\mathbf{X}'_p)$, for $c = 1, 2, ..., q$, where $\mathbf{X}_p = (\mathbf{x}_{1.,p}, \mathbf{x}_{2.,p}, ..., \mathbf{x}_{N.,p})$, $\mathbf{x}_{l.,p} = (x_{l1,p}, x_{l2,p}, ..., x_{lT,p})'$, $l = 1, 2, ..., N, p = 1, 2, ..., k, r = 0, 1, ..., R$, and $\mathbf{W}^0 \equiv \mathbf{I}_N$. It can be shown that[46]

$$\frac{1}{\sqrt{NT}}\mathbf{Q}_c'\mathbf{M}^b[(\mathbf{I}_T \otimes \boldsymbol{\Gamma}_0)\mathbf{f} + \mathbf{e}] = \frac{1}{\sqrt{NT}}\sum_{i=1}^N\sum_{l=1}^N w_{il}^{(r)}\mathbf{x}'_{l.,p}\bar{\mathbf{M}}(\mathbf{F}\boldsymbol{\gamma}_{i,0} + \mathbf{e}_{i.}),$$

where $\mathbf{F} = (\mathbf{f}_1, \mathbf{f}_2, ..., \mathbf{f}_T)'$, $\mathbf{e}_{i.} = (e_{i1}, e_{i2}, ..., e_{iT})'$, and $w_{ij}^{(r)}$ is the $(i,j)^{th}$ element of $\mathbf{W}^r$. Note that the distribution of

[46] See (S.11) in the Supporting Information for details.

$(NT)^{-1/2}\sum_{i=1}^{N}\mathbf{X}_{i.}'\bar{\mathbf{M}}(\mathbf{F}\boldsymbol{\gamma}_{i,0}+\mathbf{e}_{i.})$ can be obtained by applying Lemma S.6 and recalling that $\mathbf{X}_{i.}=\mathbf{F}\mathbf{A}_i+\mathbf{V}_{i.}$, where $\mathbf{V}_{i.}=(\mathbf{v}_{i1},\mathbf{v}_{i2},...,\mathbf{v}_{iT})'$:

$$\begin{aligned}\frac{1}{\sqrt{NT}}\sum_{i=1}^{N}\mathbf{X}_{i.}'\bar{\mathbf{M}}(\mathbf{F}\boldsymbol{\gamma}_{i,0}+\mathbf{e}_{i.}) &= \frac{1}{\sqrt{NT}}\sum_{i=1}^{N}(\mathbf{A}_i'\mathbf{F}'+\mathbf{V}_{i.}')\bar{\mathbf{M}}\mathbf{F}\boldsymbol{\gamma}_{i,0}+\frac{1}{\sqrt{NT}}\sum_{i=1}^{N}\mathbf{X}_{i.}'\bar{\mathbf{M}}\mathbf{e}_{i.}\\ &= \frac{1}{\sqrt{NT}}\sum_{i=1}^{N}\mathbf{X}_{i.}'\mathbf{M}_f\mathbf{e}_{i.}+\sqrt{\frac{T}{N}}(\boldsymbol{d}_1-\boldsymbol{d}_2-\boldsymbol{d}_3)\\ &\quad+O_p\left(\frac{\sqrt{T}}{N}\right)+O_p\left(\frac{1}{\sqrt{T}}\right)+O_p\left(\frac{1}{\sqrt{N}}\right),\end{aligned}$$

where $\boldsymbol{d}_1,\boldsymbol{d}_2$, and $\boldsymbol{d}_3$ are given by (S.3), (S.4), and (S.5) in the Supporting Information, respectively. Since Assumption 6 implies that $\mathbf{W}^r$ has bounded row and column sum norms, for $r=0,1,...,R$, we then have

$$\begin{aligned}\frac{1}{\sqrt{NT}}\mathbf{Q}'\mathbf{M}^b[(\mathbf{I}_T\otimes\boldsymbol{\Gamma})\mathbf{f}+\mathbf{e}] &= \frac{1}{\sqrt{NT}}\mathbf{Q}'\mathbf{M}_f^b\mathbf{e}+\sqrt{\frac{T}{N}}(\boldsymbol{b}_1-\boldsymbol{b}_2-\boldsymbol{b}_3)\\ &\quad+O_p\left(\frac{\sqrt{T}}{N}\right)+O_p\left(\frac{1}{\sqrt{T}}\right)+O_p\left(\frac{1}{\sqrt{N}}\right),\end{aligned}\tag{B.5}$$

where the elements of $\boldsymbol{b}_1,\boldsymbol{b}_2$, and $\boldsymbol{b}_3$ are given by (22), (23), and (24) in the main article, respectively, and then (15) in Theorem 1 readily follows as $(N,T)\xrightarrow{j}\infty$ with $\sqrt{T}/N\to 0$.

Proof of Theorem 2. Note that

$$\sqrt{NT}(\hat{\boldsymbol{\delta}}_{b2sls}-\boldsymbol{\delta}_0)=\left(\frac{1}{NT}\hat{\mathbf{Q}}^{*\prime}\mathbf{M}^b\mathbf{L}\right)^{-1}\frac{1}{\sqrt{NT}}\hat{\mathbf{Q}}^{*\prime}\mathbf{M}^b[(\mathbf{I}_T\otimes\boldsymbol{\Gamma}_0)\mathbf{f}+\mathbf{e}].$$

We first show that

$$\operatorname*{plim}_{N,T\to\infty}\frac{1}{NT}\hat{\mathbf{Q}}^{*\prime}\mathbf{M}^b\mathbf{L}=\operatorname*{plim}_{N,T\to\infty}\frac{1}{NT}\mathbf{L}_0'\mathbf{M}_f^b\mathbf{L}_0.\tag{B.6}$$

Substituting $\mathbf{Y}=(\mathbf{I}_T\otimes\mathbf{S}_0^{-1})[\mathbf{X}\boldsymbol{\beta}_0+(\mathbf{I}_T\otimes\boldsymbol{\Gamma}_0)\mathbf{f}+\mathbf{e}]$ into the definition of $\mathbf{L}$ yields $\mathbf{L}=\mathbf{L}_0+[(\mathbf{I}_T\otimes\boldsymbol{\Gamma}_0)\mathbf{f}+\mathbf{e},\mathbf{0}]$, and it follows that

$$\frac{1}{NT}\hat{\mathbf{Q}}^{*\prime}\mathbf{L}=\frac{1}{NT}\left[(\mathbf{I}_T\otimes\mathbf{G}(\hat{\rho}))\mathbf{X}\hat{\boldsymbol{\beta}},\mathbf{X}\right]'\mathbf{M}^b\mathbf{L}_0+\frac{1}{NT}\left[(\mathbf{I}_T\otimes\mathbf{G}(\hat{\rho}))\mathbf{X}\hat{\boldsymbol{\beta}},\mathbf{X}\right]'\mathbf{M}^b[(\mathbf{I}_T\otimes\boldsymbol{\Gamma}_0)\mathbf{f}+\mathbf{e},\mathbf{0}].$$

Using the first-order Taylor expansion of $\mathbf{G}(\hat{\rho})$, we have

$$\mathbf{W}(\mathbf{I}_N-\hat{\rho}\mathbf{W})^{-1}=\mathbf{G}_0+\mathbf{W}(\mathbf{I}_N-\hat{\rho}\mathbf{W})^{-1}\mathbf{G}_0(\hat{\rho}-\rho_0).$$

Applying Lemma S.7 and using $\hat{\rho}=\rho+o_p(1)$ and $\hat{\boldsymbol{\beta}}=\boldsymbol{\beta}_0+o_p(1)$, we obtain

$$\begin{aligned}\frac{1}{NT}\left[(\mathbf{I}_T\otimes\mathbf{G}(\hat{\rho}))\mathbf{X}\hat{\boldsymbol{\beta}},\mathbf{X}\right]'\mathbf{M}^b\mathbf{L}_0&=\frac{1}{NT}\mathbf{L}_0'\mathbf{M}_f^b\mathbf{L}_0+o_p(1),\\ \frac{1}{NT}\left[(\mathbf{I}_T\otimes\mathbf{G}(\hat{\rho}))\mathbf{X}\hat{\boldsymbol{\beta}},\mathbf{X}\right]'\mathbf{M}^b[(\mathbf{I}_T\otimes\boldsymbol{\Gamma}_0)\mathbf{f}+\mathbf{e},\mathbf{0}]&=o_p(1),\end{aligned}$$

and hence, the result in (B.6) is proved. By similar reasoning as in the proof of Theorem 1, we have

$$\begin{aligned}\frac{1}{\sqrt{NT}}\hat{\mathbf{Q}}^{*\prime}\mathbf{M}^b[(\mathbf{I}_T\otimes\boldsymbol{\Gamma}_0)\mathbf{f}+\mathbf{e}] &= \frac{1}{\sqrt{NT}}\sum_{i=1}^{N}\mathbf{L}_0'\mathbf{M}_f^b\mathbf{e}+\sqrt{\frac{T}{N}}\boldsymbol{b}_{2sls}\\ &\quad+O_p\left(\frac{\sqrt{T}}{N}\right)+O_p\left(\frac{1}{\sqrt{T}}\right)+O_p\left(\frac{1}{\sqrt{N}}\right),\end{aligned}$$

where $\boldsymbol{b}_{2sls}$ is given by (32) in the main article. Therefore, the result in (30) readily follows as $(N,T)\xrightarrow{j}\infty$ with $\sqrt{T}/N\to 0$.

When $T/N\to 0$ as $(N,T)\xrightarrow{j}\infty$, both 2SLS and B2SLS estimators are asymptotically unbiased. To examine if $\hat{\mathbf{Q}}^*$ is the best IV, we just need to compare the asymptotic variances $\boldsymbol{\Sigma}_{b2sls}$ with $\boldsymbol{\Sigma}_{2sls}$. Notice that $\mathbf{L}_0'\mathbf{P}_{Q,f}\mathbf{L}_0=\mathbf{L}_0'\mathbf{M}_f^b\mathbf{Q}(\mathbf{Q}'\mathbf{M}_f^b\mathbf{Q})^{-1}\mathbf{Q}'\mathbf{M}_f^b\mathbf{L}_0\le\mathbf{L}_0'\mathbf{M}_f^b\mathbf{L}_0$, and hence $\boldsymbol{\Psi}_{LPL}\le\boldsymbol{\Psi}_{LML}$. If the errors $\{e_{it}\}$ are independently and identically distributed with mean zero and variance σ_e^2, then $\boldsymbol{\Sigma}_{b2sls}=\sigma_e^2\boldsymbol{\Psi}_{LML}^{-1}\le\sigma_e^2\boldsymbol{\Psi}_{LPL}^{-1}=\boldsymbol{\Sigma}_{2sls}$. However, in general we cannot conclude that $\hat{\mathbf{Q}}^*$ is optimal, because $\boldsymbol{\Omega}_{e,i.}$ is unknown and $\boldsymbol{\Omega}_{LPe}$ could be greater than $\boldsymbol{\Omega}_{LMe}$.

Proof of Theorem 3.

Consistency: We first show that $\mathbf{A}_{NT}^{w}\mathbf{g}_{NT}(\boldsymbol{\delta})$ converges to its mean uniformly in $\boldsymbol{\delta}\in\Delta_{sp}$ and the limit equals zero at $\boldsymbol{\delta}_0$. Consider $\boldsymbol{\xi}(\boldsymbol{\delta})$ given by (34) in the main article. It can be rewritten as

$$\boldsymbol{\xi}(\boldsymbol{\delta})=[\mathbf{I}_T\otimes\mathbf{S}(\rho)]\left(\mathbf{I}_T\otimes\mathbf{S}_0^{-1}\right)[\mathbf{X}\boldsymbol{\beta}_0+(\mathbf{I}_T\otimes\boldsymbol{\Gamma}_0)\mathbf{f}+\mathbf{e}]-\mathbf{X}\boldsymbol{\beta}.$$

Since $\mathbf{S}(\rho)\mathbf{S}_0^{-1}=[\mathbf{S}_0+(\rho_0-\rho)\mathbf{W}]\mathbf{S}_0^{-1}=\mathbf{I}_N+(\rho_0-\rho)\mathbf{G}_0$, we obtain

$$\begin{aligned}\boldsymbol{\xi}(\boldsymbol{\delta})=&[\mathbf{I}_{NT}+(\rho_0-\rho)(\mathbf{I}_T\otimes\mathbf{G}_0)][\mathbf{X}\boldsymbol{\beta}_0+(\mathbf{I}_T\otimes\boldsymbol{\Gamma}_0)\mathbf{f}]-\mathbf{X}\boldsymbol{\beta}+\mathbf{I}_T\otimes\left[\mathbf{S}(\rho)\mathbf{S}_0^{-1}\right]\mathbf{e}\\=&\mathbf{d}(\boldsymbol{\delta})+\mathbf{r}_{\xi}(\rho).\end{aligned}\tag{B.7}$$

where $\mathbf{d}(\boldsymbol{\delta})=\mathbf{X}(\boldsymbol{\beta}_0-\boldsymbol{\beta})+(\rho_0-\rho)(\mathbf{I}_T\otimes\mathbf{G}_0)[\mathbf{X}\boldsymbol{\beta}_0+(\mathbf{I}_T\otimes\boldsymbol{\Gamma}_0)\mathbf{f}]$ and $\mathbf{r}_{\xi}(\rho)=(\mathbf{I}_T\otimes\boldsymbol{\Gamma}_0)\mathbf{f}+\mathbf{I}_T\otimes\left[\mathbf{S}(\rho)\mathbf{S}_0^{-1}\right]\mathbf{e}$. Let $\mathbf{A}_{NT}^{w}=(\mathbf{a}_1^{w},\mathbf{a}_2^{w},...,\mathbf{a}_h^{w},\mathbf{A}_Q^{w})$, where $\mathbf{a}_l^{w}$ are $k_a\times 1$ vectors, for $l=1,2,...,h$, and $\mathbf{A}_Q^{w}$ is a $k_a\times q$ matrix. By definition, we have

$$\mathbf{A}_{NT}^{w}\mathbf{g}_{NT}(\boldsymbol{\delta})=\frac{1}{NT}\sum_{l=1}^{h}\mathbf{a}_l^{w}\left[\boldsymbol{\xi}'(\boldsymbol{\delta})\mathbf{M}^b\mathbf{P}_l^b\mathbf{M}^b\boldsymbol{\xi}(\boldsymbol{\delta})\right]+\frac{1}{NT}\mathbf{A}_Q^{w}\mathbf{Q}'\mathbf{M}^b\boldsymbol{\xi}(\boldsymbol{\delta}).\tag{B.8}$$

Expanding the first term on the right-hand side of (B.8) produces

$$\frac{1}{NT}\sum_{l=1}^{h}\mathbf{a}_l^{w}\left[\boldsymbol{\xi}'(\boldsymbol{\delta})\mathbf{M}^b\mathbf{P}_l^b\mathbf{M}^b\boldsymbol{\xi}(\boldsymbol{\delta})\right]=\boldsymbol{\varpi}_1+2\boldsymbol{\varpi}_2+\boldsymbol{\varpi}_3,$$

where $\boldsymbol{\varpi}_1=\frac{1}{NT}\sum_{l=1}^{h}\mathbf{a}_l^{w}\left[\mathbf{d}'(\boldsymbol{\delta})\mathbf{M}^b\mathbf{P}_l^b\mathbf{M}^b\mathbf{d}(\boldsymbol{\delta})\right]$, $\boldsymbol{\varpi}_2=\frac{1}{NT}\sum_{l=1}^{h}\mathbf{a}_l^{w}\left[\mathbf{d}'(\boldsymbol{\delta})\mathbf{M}^b\mathbf{P}_l^b\mathbf{M}^b\mathbf{r}_{\xi}(\rho)\right]$, and $\boldsymbol{\varpi}_3=\frac{1}{NT}\sum_{l=1}^{h}\mathbf{a}_l^{w}\left[\mathbf{r}_{\xi}'(\rho)\mathbf{M}^b\mathbf{P}_l^b\mathbf{M}^b\mathbf{r}_{\xi}(\rho)\right]$. Note that $\mathbf{S}(\rho)\mathbf{S}_0^{-1}$ has bounded row and column norms, and so do the products of $\mathbf{S}(\rho)\mathbf{S}_0^{-1}$, $\mathbf{P}_l$, and $\left[\mathbf{S}(\rho)\mathbf{S}_0^{-1}\right]'$. Also notice that $\mathbf{M}^b\mathbf{P}_l^b\mathbf{M}^b=\bar{\mathbf{M}}\otimes\mathbf{P}_l$. Applying Lemmas S.7 and S.8, we obtain that $\boldsymbol{\varpi}_1,\boldsymbol{\varpi}_2$, and $\boldsymbol{\varpi}_3$ converge uniformly to their means, respectively. Consider now the second term on the right-hand side of (B.8). Using (B.7), we have

$$\begin{aligned}\frac{1}{NT}\mathbf{Q}'\mathbf{M}^b\boldsymbol{\xi}(\boldsymbol{\delta})=&\frac{1}{NT}\mathbf{Q}'\mathbf{M}^b\mathbf{d}(\boldsymbol{\delta})+\frac{1}{NT}\mathbf{Q}'\mathbf{M}^b\mathbf{r}_{\xi}(\rho)\\=&-\frac{1}{NT}\mathbf{Q}'\mathbf{M}^b\mathbf{L}_0(\boldsymbol{\delta}-\boldsymbol{\delta}_0)-\frac{1}{NT}(\rho-\rho_0)\mathbf{Q}'\mathbf{M}^b(\mathbf{I}_T\otimes\mathbf{G}_0\boldsymbol{\Gamma}_0)\mathbf{f}\\&+\frac{1}{NT}\mathbf{Q}'\mathbf{M}^b(\mathbf{I}_T\otimes\boldsymbol{\Gamma}_0)\mathbf{f}+\frac{1}{NT}\mathbf{Q}'\mathbf{M}^b\left\{\mathbf{I}_T\otimes\left[\mathbf{S}(\rho)\mathbf{S}_0^{-1}\right]\mathbf{e}\right\}\\=&-\frac{1}{NT}\mathbf{Q}'\mathbf{M}_f^b\mathbf{L}_0(\boldsymbol{\delta}-\boldsymbol{\delta}_0)+O_p\left(\frac{1}{N}\right)+O_p\left(\frac{1}{\sqrt{NT}}\right),\end{aligned}\tag{B.9}$$

where the last equality follows by Lemma S.7. Under Assumption 8(a), the limit of $(NT)^{-1}\mathbf{Q}'\mathbf{M}_f^b\mathbf{L}_0$ exists and then the uniform convergence of $(NT)^{-1}\mathbf{Q}'\mathbf{M}^b\boldsymbol{\xi}(\boldsymbol{\delta})$ follows. Next, we verify that the limit of $\mathbf{A}_{NT}^{w}\mathbf{g}_{NT}(\boldsymbol{\delta})$ equals zero at the true parameter value $\boldsymbol{\delta}_0$. Noting that $\boldsymbol{\xi}(\boldsymbol{\delta}_0)=(\mathbf{I}_T\otimes\boldsymbol{\Gamma}_0)\mathbf{f}+\mathbf{e}$, we have

$$\begin{aligned}\frac{1}{NT}E\left(\mathbf{e}'\mathbf{M}_f^b\mathbf{P}_l^b\mathbf{M}_f^b\mathbf{e}\right)=&\frac{1}{NT}\sum_{i=1}^{N}\sum_{j=1}^{N}E\left(\mathbf{e}_{i.}'\mathbf{M}_f\mathbf{e}_{j.}\right)p_{l,ji}\\=&\frac{1}{NT}\sum_{i=1}^{N}E\left(\mathbf{e}_{i.}'\mathbf{M}_f\mathbf{e}_{i.}\right)p_{l,ii}+\frac{1}{NT}\sum_{i=1}^{N}\sum_{j=1,j\neq i}^{N}E\left(\mathbf{e}_{i.}'\mathbf{M}_f\mathbf{e}_{j.}\right)p_{l,ji}\\=&0+\frac{1}{NT}\sum_{i=1}^{N}\sum_{j=1,j\neq i}^{N}p_{l,ji}E\left\{tr\left[\mathbf{M}_fE\left(\mathbf{e}_{j.}\mathbf{e}_{i.}'\right)\right]|\mathbf{F}\right\}=0,\end{aligned}$$

and $(NT)^{-1}\mathbf{Q}'\mathbf{M}^b\boldsymbol{\xi}(\boldsymbol{\delta}_0)=\mathbf{0}$ can be readily checked from (B.9). Finally, we need to show that $\operatorname{plim}_{N\to\infty}\mathbf{A}_{NT}^{w}\mathbf{g}_{NT}(\boldsymbol{\delta})=\mathbf{0}$ has a unique solution at $\boldsymbol{\delta}_0$. Since $(NT)^{-1}\mathbf{Q}'\mathbf{M}_f^b\mathbf{L}_0$ has full column rank for large enough N under Assumption 8(a), from (B.9) it can be seen that $\boldsymbol{\delta}_0$ can be identified via the linear moment conditions. In sum, the consistency of the estimator follows from both uniform convergence of $\mathbf{A}_{NT}^{w}\mathbf{g}_{NT}(\boldsymbol{\delta})$ and identifiability of $\boldsymbol{\delta}_0$.

Asymptotic distribution: By a mean value expansion of $\frac{\partial\mathbf{g}_{NT}'(\hat{\boldsymbol{\delta}}_{gmm})}{\partial}\mathbf{A}_{NT}^{w\prime}\mathbf{A}_{NT}^{w}\mathbf{g}_{NT}(\hat{\boldsymbol{\delta}}_{gmm})=0$ around the true value, $\boldsymbol{\delta}_0$, we obtain

$$\sqrt{NT}\left(\hat{\boldsymbol{\delta}}_{gmm}-\boldsymbol{\delta}_0\right)=-\left[\frac{\partial \mathbf{g}_{NT}'\left(\hat{\boldsymbol{\delta}}_{gmm}\right)}{\partial \boldsymbol{\delta}}\mathbf{A}_{NT}^{w\prime}\mathbf{A}_{NT}^{w}\frac{\partial \mathbf{g}_{NT}(\ddot{\boldsymbol{\delta}})}{\partial \boldsymbol{\delta}'}\right]^{-1}\frac{\partial \mathbf{g}_{NT}'(\hat{\boldsymbol{\delta}})}{\partial \boldsymbol{\delta}}\mathbf{A}_{NT}^{w\prime}\sqrt{NT}\mathbf{A}_{NT}^{w}\mathbf{g}_{NT}(\boldsymbol{\delta}_0),$$

where $\ddot{\boldsymbol{\delta}}$ lies element by element between $\hat{\boldsymbol{\delta}}$ and $\boldsymbol{\delta}_0$. For any $\boldsymbol{\delta}$ in the parameter space Δ_{sp}, we have $\partial\boldsymbol{\xi}(\boldsymbol{\delta})/\partial\boldsymbol{\delta}'=-\mathbf{L}=-(\mathbf{Y}^*,\mathbf{X})$, and it follows that

$$\frac{\partial \mathbf{g}_{NT}(\boldsymbol{\delta})}{\partial \boldsymbol{\delta}'}=-\frac{1}{NT}\left[\mathbf{M}^b\mathbf{P}_1^{sb}\mathbf{M}^b\boldsymbol{\xi}(\boldsymbol{\delta}),\ldots,\mathbf{M}^b\mathbf{P}_h^{sb}\mathbf{M}^b\boldsymbol{\xi}(\boldsymbol{\delta}),\mathbf{M}^b\mathbf{Q}\right]'(\mathbf{Y}^*,\mathbf{X}),$$

where $\mathbf{P}_l^{sb}=\mathbf{I}_T\otimes\mathbf{P}_l^s$ and $\mathbf{P}_l^s=\mathbf{P}_l+\mathbf{P}_l'$, for $l=1,2,\ldots,h$. Since, $\mathbf{Y}^*=(\mathbf{I}_T\otimes\mathbf{W})\mathbf{Y}$, it is easily seen that

$$\begin{aligned}\frac{1}{NT}\boldsymbol{\xi}'(\boldsymbol{\delta})\mathbf{M}^b\mathbf{P}_l^{sb}\mathbf{M}^b\mathbf{Y}^*=&\frac{1}{NT}\boldsymbol{\xi}'(\boldsymbol{\delta})\left(\bar{\mathbf{M}}\otimes\mathbf{P}_l^s\mathbf{G}_0\right)\mathbf{X}\boldsymbol{\beta}_0\\&+\frac{1}{NT}\boldsymbol{\xi}'(\boldsymbol{\delta})\left[\left(\bar{\mathbf{M}}\otimes\mathbf{P}_l^s\mathbf{G}_0\right)\mathbf{e}+\left(\bar{\mathbf{M}}\otimes\mathbf{P}_l^s\mathbf{G}_0\boldsymbol{\Gamma}_0\right)\mathbf{f}\right].\end{aligned}\tag{B.10}$$

By Lemmas S.7 and S.8, at true value $\boldsymbol{\delta}_0$ (B.10) can be rewritten as

$$\frac{1}{NT}[(\mathbf{I}_T\otimes\boldsymbol{\Gamma}_0)\mathbf{f}+\mathbf{e}]'\mathbf{M}^b\mathbf{P}_l^{sb}\mathbf{M}^b\mathbf{Y}^*=\frac{1}{N}\sum_{i=1}^{N}\tilde{g}_{l,ii}^s\sigma_i^2+o_p(1),$$

where $\tilde{g}_{l,ii}^s$ is the ith diagonal element of matrix $\tilde{\mathbf{G}}_l(\rho_0)=\mathbf{P}_l^s\mathbf{G}_0$, and $(NT)^{-1}\mathbf{e}'\mathbf{M}^b\mathbf{P}_l^s\mathbf{M}^b\mathbf{X}=o_p(1)$. In addition, we have shown in (B.4) that $(NT)^{-1}\mathbf{Q}'\mathbf{M}^b\mathbf{L}=(NT)^{-1}\mathbf{Q}'\mathbf{M}_f^b\mathbf{L}_0+o_p(1)$. It then follows that $\partial\mathbf{g}_{NT}'(\boldsymbol{\delta})/\partial\boldsymbol{\delta}=-\mathbf{D}+o_p(1)$, where $\mathbf{D}$ is given by (38). Finally, turning to $\sqrt{NT}\mathbf{A}_{NT}^w\mathbf{g}_{NT}(\boldsymbol{\delta}_0)$. Applying Lemma S.8 and (B.5) yields

$$\begin{aligned}\sqrt{NT}\mathbf{A}_{NT}^w\mathbf{g}_{NT}(\boldsymbol{\delta}_0)=&\frac{1}{\sqrt{NT}}\left[\left(\sum_{l=1}^{h}\mathbf{a}_l^w\mathbf{r}_{\xi}'(\rho_0)\mathbf{M}^b\mathbf{P}_l^b\mathbf{M}^b\mathbf{r}_{\xi}(\rho_0)\right)+\mathbf{A}_Q^w\mathbf{Q}'\mathbf{M}^b\mathbf{r}_{\xi}(\rho_0)\right]\\=&\frac{1}{\sqrt{NT}}\sum_{l=1}^{h}\mathbf{a}_l^w[(\mathbf{I}_T\otimes\boldsymbol{\Gamma}_0)\mathbf{f}+\mathbf{e}]'\left(\mathbf{M}^b\mathbf{P}_l^b\mathbf{M}^b\right)[(\mathbf{I}_T\otimes\boldsymbol{\Gamma}_0)\mathbf{f}+\mathbf{e}]\\&+\frac{1}{\sqrt{NT}}\mathbf{A}_Q^w\mathbf{Q}'\mathbf{M}^b[(\mathbf{I}_T\otimes\boldsymbol{\Gamma}_0)\mathbf{f}+\mathbf{e}]\\=&\frac{1}{\sqrt{NT}}\left[\sum_{l=1}^{h}\mathbf{a}_l^w\mathbf{e}'(\bar{\mathbf{M}}\otimes\mathbf{P}_l)\mathbf{e}+\mathbf{A}_Q^w\sum\mathbf{Q}'\mathbf{M}_f^b\mathbf{e}\right]+\mathbf{A}^w\sqrt{\frac{T}{N}}\boldsymbol{b}_{gmm}\\&+O_p\left(\frac{\sqrt{T}}{N}\right)+O_p\left(\frac{1}{\sqrt{T}}\right)+O_p\left(\frac{1}{\sqrt{N}}\right),\end{aligned}$$

where $\boldsymbol{b}_{gmm}$ is given by (39). Then, applying the central limit theorem given by Lemma S.10 for the linear and quadratic forms establishes

$$\frac{1}{\sqrt{NT}}\left[\sum_{l=1}^{h}\mathbf{a}_l^w\mathbf{e}'(\mathbf{I}_T\otimes\mathbf{P}_l)\mathbf{e}+\mathbf{A}_Q^w\sum\mathbf{Q}'\mathbf{M}_f^b\mathbf{e}\right]\sim N\left(\mathbf{0},\mathbf{A}^w\boldsymbol{\Sigma}_g\mathbf{A}^{w\prime}\right),$$

where $\boldsymbol{\Sigma}_g$ is given by (38), and this completes the proof.

Appendix C: Data Appendix

The house price indices for MSAs at monthly frequency are obtained from the website of Freddie Mac. The quarterly values are computed by taking the three-month arithmetic averages. The annual CPI series for all urban areas is sourced from website of the Bureau of Labor Statistics. The CPI for each MSA is constructed from the corresponding state CPI, and the missing observations for a few area-year combinations are replaced by the US averages. The data on annual income per capita and population at the MSA level are obtained from the website of the BEA. The quarterly values of CPI, income, and population are computed from annual series, following the interpolation method given in Appendix B.3 of the Global Vector AutoRegressive (GVAR) Toolbox User Guide, which is available at the GVAR modeling website: https://sites.google.com/site/gvarmodelling/gvar-toolbox/download.

The geodesic distance between MSAs is calculated by the Haversine formula, using the Latitude-Longitude of zip codes of the corresponding MSAs. The data on MSA-to-MSA migration flows are sourced from the 2010 to

Table C1. Summary of the spatial weights matrices.

	$\mathbf{W}_{75}$	$\mathbf{W}_{100}$	$\mathbf{W}_{125}$	$\mathbf{W}_{m}$	$\hat{\mathbf{W}}^{+}$	$\hat{\mathbf{W}}^{-}$
Links						
Mean	3.31	5.73	8.65	4.46	10.41	7.53
Max	12	20	27	59	35	35
Total	1,246	2,162	3,260	1,681	3,926	2,838
Network density	0.88%	1.53%	2.30%	1.19%	2.77%	2.00%
Isolated MSAs	39	15	10	53	3	45
Dimensions	377×377					

Notes: $\mathbf{W}_d$ denotes radial distance weights matrix with threshold distance d (miles). $\mathbf{W}_m$ represents weights matrix based on MSA-to-MSA migration flows. $\hat{\mathbf{W}}^{+}$ ($\hat{\mathbf{W}}^{-}$) is constructed from significantly positive (negative) pairwise correlations of de-factored house price changes. The total number of links equals the number of nonzero elements in the weights matrix. Network density is computed by dividing the sum of existing links by the number of all possible links.

2014 American Community Survey (ACS) 5-year estimates by the United States Census Bureau. The flow estimates with coefficients of variation higher than 20% are dropped from the sample. Table C1 reports the summary measures of different spatial weights matrices used in the analysis. Further details about these weights matrices are provided in the Supporting Information.

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